

PDE Ledger  
Archive Ledger Skeleton

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# Purpose and Scope

**TODO.** B3 will fill the rebuilt ledger purpose, source hierarchy, and scope boundaries.

## 0.1 The Role of This Ledger

Placeholder for the rebuilt ledger's role and audience.

## 0.2 Primary Source Hierarchy

Placeholder for the source hierarchy used by the rebuilt ledger.

## 0.3 Document Contract

Placeholder for the provenance, status, verification, and downstream-use contract.

## 0.4 Scope Boundaries

Placeholder for the limits of the rebuilt ledger.

## 0.5 Completeness Criterion

Placeholder for the completion standard used by later build phases.

# How to Read This Ledger

**TODO.** B3 will fill the reader guide after the rebuilt stage and Part structure is set.

## 0.6 Layer Structure

Placeholder for the relationship between theorem-block prose, stage cards, notes, and audits.

## 0.7 Reading Paths

Placeholder for operational, verification, provenance, and authoring reading paths.

## 0.8 Citation Discipline

Placeholder for result anchors, theorem labels, equation labels, and stage provenance.

## 0.9 Stage Entries

Placeholder for the expected fields in future stage cards.

## 0.10 Gap Handling

Placeholder for demotion, splitting, open-gate, and failed-route handling.

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# Claim-Status Firewall

**TODO.** B3 will fill the rebuilt ledger's status rules and examples.

## 0.11 Allowed Status Tags

| Status tag                        | Meaning      |
|-----------------------------------|--------------|
| EXACT                             | Placeholder. |
| EXACT WITHIN CLOSURE              | Placeholder. |
| REDUCED / CONTROLLED<br>REDUCTION | Placeholder. |
| EFFECTIVE CLOSURE                 | Placeholder. |
| NUMERICALLY LOCATED               | Placeholder. |
| OPEN                              | Placeholder. |

## 0.12 Audit-Status Vocabulary

Placeholder for audit evidence categories.

## 0.13 Status Inheritance

Placeholder for weakest-essential-input and status propagation rules.

## 0.14 Citation Language

Placeholder for safe status-aware citation wording.

# Notation Firewall

**TODO.** B3 will fill notation rules after the rebuilt sectors and stage packets are fixed.

## 0.15 Global Anti-Ambiguity Rules

Placeholder for global notation safeguards.

## 0.16 Coordinate, Index, and Operator Conventions

Placeholder for coordinate, index, and operator conventions.

## 0.17 Field and Sector Variables

Placeholder for sector-specific variable conventions.

## 0.18 High-Risk Symbol Collision Table

Placeholder for symbol collision accounting.

## 0.19 Notation Checklist

Placeholder for future stage write-up checks.

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# Result Anchor Map

**TODO.** B3 will assign rebuilt result anchors after the Part and stage plan is settled.

## 0.20 Anchor Use Rules

Placeholder for stable citation-anchor rules.

## 0.21 Anchor Naming Convention

Placeholder for rebuilt anchor naming.

## 0.22 Top-Level Result Anchors

| Anchor                   | Stage range | Citation target |
|--------------------------|-------------|-----------------|
| No anchors assigned yet. |             |                 |

## 0.23 Maintenance Checklist

Placeholder for anchor maintenance rules.

# Dependency Map

**TODO.** B3 will fill the rebuilt dependency graph after stages are authored.

## 0.24 Arrow Types

Placeholder for dependency-arrow types.

## 0.25 Global Dependency Spine

Placeholder for the rebuilt ledger's dependency spine.

## 0.26 Top-Level Dependency Table

| Block                              | Inputs | Outputs |
|------------------------------------|--------|---------|
| No dependency blocks assigned yet. |        |         |

## 0.27 Revision Rules

Placeholder for dependency-map revision rules.

# Contents

|                                                            |            |
|------------------------------------------------------------|------------|
| <b>Purpose and Scope</b>                                   | <b>i</b>   |
| 0.1 The Role of This Ledger . . . . .                      | i          |
| 0.2 Primary Source Hierarchy . . . . .                     | i          |
| 0.3 Document Contract . . . . .                            | i          |
| 0.4 Scope Boundaries . . . . .                             | i          |
| 0.5 Completeness Criterion . . . . .                       | i          |
| <b>How to Read This Ledger</b>                             | <b>ii</b>  |
| 0.6 Layer Structure . . . . .                              | ii         |
| 0.7 Reading Paths . . . . .                                | ii         |
| 0.8 Citation Discipline . . . . .                          | ii         |
| 0.9 Stage Entries . . . . .                                | ii         |
| 0.10 Gap Handling . . . . .                                | ii         |
| <b>Claim-Status Firewall</b>                               | <b>iii</b> |
| 0.11 Allowed Status Tags . . . . .                         | iii        |
| 0.12 Audit-Status Vocabulary . . . . .                     | iii        |
| 0.13 Status Inheritance . . . . .                          | iii        |
| 0.14 Citation Language . . . . .                           | iii        |
| <b>Notation Firewall</b>                                   | <b>iv</b>  |
| 0.15 Global Anti-Ambiguity Rules . . . . .                 | iv         |
| 0.16 Coordinate, Index, and Operator Conventions . . . . . | iv         |
| 0.17 Field and Sector Variables . . . . .                  | iv         |
| 0.18 High-Risk Symbol Collision Table . . . . .            | iv         |
| 0.19 Notation Checklist . . . . .                          | iv         |
| <b>Result Anchor Map</b>                                   | <b>v</b>   |
| 0.20 Anchor Use Rules . . . . .                            | v          |
| 0.21 Anchor Naming Convention . . . . .                    | v          |
| 0.22 Top-Level Result Anchors . . . . .                    | v          |
| 0.23 Maintenance Checklist . . . . .                       | v          |
| <b>Dependency Map</b>                                      | <b>vi</b>  |
| 0.24 Arrow Types . . . . .                                 | vi         |
| 0.25 Global Dependency Spine . . . . .                     | vi         |
| 0.26 Top-Level Dependency Table . . . . .                  | vi         |
| 0.27 Revision Rules . . . . .                              | vi         |



|            |                                                                                                                                                 |           |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>I</b>   | <b>Part I – The medium</b>                                                                                                                      | <b>1</b>  |
| 1          | Medium Ledger Stages                                                                                                                            | 2         |
| <b>II</b>  | <b>Part II – Gravity</b>                                                                                                                        | <b>3</b>  |
| 2          | Gravity Ledger Stages                                                                                                                           | 4         |
| <b>III</b> | <b>Part III – Light</b>                                                                                                                         | <b>5</b>  |
| 3          | Light Ledger Stages                                                                                                                             | 6         |
| <b>A</b>   | <b>Part I Stage Cards</b>                                                                                                                       | <b>7</b>  |
| A.1        | Stage 004: GNLS Action and Dimensional Foundation . . . . .                                                                                     | 7         |
| A.2        | Stage 005: EOS Sound Speed and Light/Sound Ratio as a Free Knob . . . . .                                                                       | 8         |
| A.3        | Stage 006: Two-Phase Material-State Ontology (Order Field $\chi_B$ ) . . . . .                                                                  | 9         |
| A.4        | Stage 007: Shear-Surface G0 Freeze (Frozen Medium Action, Drift Ledger, DOF) . .                                                                | 11        |
| <b>B</b>   | <b>Part II Stage Cards</b>                                                                                                                      | <b>13</b> |
| B.1        | Stage 001: Solid-Angle and Second-Moment Primitives . . . . .                                                                                   | 13        |
| B.2        | Stage 002: Matter-Stress Inter-Defect Force Assembly . . . . .                                                                                  | 14        |
| B.3        | Stage 008: Monopole/Dipole Return Constraint Specification . . . . .                                                                            | 15        |
| B.4        | Stage 009: Flat-Slab Return Residual (Check A) . . . . .                                                                                        | 16        |
| B.5        | Stage 010: Slab Localization $p = 2$ with Reachable NOGO (Check B) . . . . .                                                                    | 17        |
| B.6        | Stage 011: Frozen-Reduction Certificate ( $L_s$ Helmholtz) (Check II-G1a) . . . . .                                                             | 18        |
| B.7        | Stage 012: DtN Pole Ladder & Robin Falsifier ( $Z_{00} = -(\omega/c_S) \tan$ ) (Check II-G1b) .                                                 | 20        |
| B.8        | Stage 013: Breathing Harmonic Profiles & $M/K$ Operator Projection (Check II-G2a)                                                               | 22        |
| B.9        | Stage 014: Truncation Consistency — Combined-Basis Generalized Eigenproblem,<br>$\beta_{L_0}$ Sweep, and N-Convergence (Check II-G2b) . . . . . | 24        |
| B.10       | Stage 015: Legacy-Hessian Structure Recovery & the Hellmann–Feynman Force<br>(Check II-G2c) . . . . .                                           | 26        |
| B.11       | Stage 016: The $\ell = 2$ SO(3) Covariance Theorem (Check II-G3a) . . . . .                                                                     | 28        |
| B.12       | Stage 017: Grouped-P2 Lane Isotropy (Check II-G3b) . . . . .                                                                                    | 30        |
| B.13       | Stage 018: The Outgoing $\ell = 2$ DtN Hankel Fingerprint and $\chi_Q$ Sign (Check II-G4a)                                                      | 32        |
| B.14       | Stage 019: The Squared-Denominator Prefactor Algebra $P(\omega) = D_0 N / D^{\text{cons}^2}$<br>(Check II-G4b) . . . . .                        | 34        |
| B.15       | Stage 020: The $54/5 = 2 \cdot 27/5$ Provenance Partition and the CALIBRATED Verdict<br>Label (Check II-G4c) . . . . .                          | 36        |
| B.16       | Stage 021: The $\hat{\mu}_0$ -free $[P_0^{\text{phys}}] = 1$ Dimensional Closure (Check II-G4d) . . . . .                                       | 39        |
| B.17       | Stage 022: The Cross- $\ell - (\ell + 1)/\Lambda_\ell$ Fingerprints + the Gate-4 Non-Regression<br>(Check II-G5a) . . . . .                     | 42        |
| <b>C</b>   | <b>Part III Stage Cards</b>                                                                                                                     | <b>45</b> |
| C.1        | Stage 003: Transverse Photon Pair and the Stray Longitudinal Mode . . . . .                                                                     | 45        |

|                                                         |           |
|---------------------------------------------------------|-----------|
| <b>D Reproducibility Map</b>                            | <b>47</b> |
| D.1 Purpose of This Appendix . . . . .                  | 47        |
| D.2 Reproducibility Layers . . . . .                    | 47        |
| D.3 Canonical Source Registry . . . . .                 | 47        |
| D.4 Primary Symbolic-Audit Registry . . . . .           | 47        |
| D.5 Repository Freeze Protocol . . . . .                | 47        |
| <b>E Source File Index</b>                              | <b>48</b> |
| E.1 Purpose of This Appendix . . . . .                  | 48        |
| E.2 Source Classes . . . . .                            | 48        |
| E.3 Generated Ledger Files . . . . .                    | 48        |
| E.4 Precedence and Conflict-Resolution Rules . . . . .  | 48        |
| <b>F Layer-2 Fit-Insertion Index</b>                    | <b>49</b> |
| F.1 Generated Coverage Summary . . . . .                | 49        |
| F.2 Phase-A Candidate Rows . . . . .                    | 49        |
| F.3 Visible Grouping and Exclusion Accounting . . . . . | 49        |
| <b>G Layer-2 Parameter-Provenance Audit</b>             | <b>50</b> |
| G.1 Generated Coverage Summary . . . . .                | 50        |
| G.2 Published-Target Benchmark Sources . . . . .        | 50        |
| G.3 Parameter Genealogy Rows . . . . .                  | 50        |
| G.4 Visible Alias Accounting . . . . .                  | 50        |
| <b>H Fill Workflow and Editorial Rules</b>              | <b>51</b> |
| H.1 Purpose of This Appendix . . . . .                  | 51        |
| H.2 What Filled Means . . . . .                         | 51        |
| H.3 Governing Maps Used During Filling . . . . .        | 51        |
| H.4 Stage Appendix Fill Protocol . . . . .              | 51        |
| H.5 Theorem-Block Fill Protocol . . . . .               | 51        |
| H.6 Minimal Build and Static-Check Workflow . . . . .   | 51        |

## Part I

# Part I – The medium

## Chapter 1

# Medium Ledger Stages

Stage cards for Part I are collected in Appendix A.

## Part II

### Part II – Gravity

## Chapter 2

# Gravity Ledger Stages

Stage cards for Part II are collected in Appendix B.

## Part III

# Part III – Light

## Chapter 3

# Light Ledger Stages

Stage cards for Part III are collected in Appendix C.



# Appendix A

## Part I Stage Cards

### A.1 Stage 004: GNLS Action and Dimensional Foundation

**Part / anchor.** Part I – The medium

**Claim status.** EXACT WITHIN CLOSURE

**Purpose.** Fix the field content and  $\{L, T, M\}$  dimensional dictionary of the GNLS parent action, derive the pin null-relation and healing scale, and land the base-system verdict `RETAIN_L_T_M`.

**Status.** EXACT WITHIN CLOSURE. The dimensional dictionary, sound-speed and healing anchors, pin null-relation, and base-system verdict are exact symbolic results; the honest landing carries named residuals (mass non-emergence and the  $\{L, T\}$ -conversion gaps), deferred to pathA\_21/22 and not repaired here.

**Parent action and field content.** The 4D-bulk medium is a GNLS condensate with order parameter  $\psi$ , number density  $\rho = |\psi|^2$ , and EOS  $P = K\rho^5$  (from  $U(\rho) = \frac{K}{4}\rho^5$ ):

$$\mathcal{L} \sim i\hbar \psi^* \partial_t \psi - \frac{\hbar^2}{2m_{\text{GNLS}}} |\nabla \psi|^2 - U(\rho).$$

**Dimensional dictionary (two tiers).** **Primitive inputs.**  $[\hbar] = ML^2T^{-1}$ ,  $[m_{\text{GNLS}}] = M$ ,  $[\rho_0] = L^{-4}$  (state datum) — posted and consistency-checked against action usage. **Derived by composition.**  $[\psi] = L^{-2}$ ,  $[\rho] = L^{-4}$ ,  $[K] = ML^{18}T^{-2}$ , forced by action homogeneity across all 17 harness checks (14 foundation, including the bulk continuity equation, plus 3  $\{L, T\}$ -representation gates).

**Derived anchors.** Sound speed and healing length.

$$c_{s0}^2 = \frac{5K\rho_0^4}{m_{\text{GNLS}}}, \quad [c_{s0}] = LT^{-1}; \quad h_0 = \frac{5K}{4}\rho_0^4 = \frac{m_{\text{GNLS}}c_{s0}^2}{4}, \quad \xi_h = \frac{\sqrt{2}\hbar}{m_{\text{GNLS}}c_{s0}}.$$

**Pin null-relation.** Four pins  $\{a, c_{s0}, \hbar, m_{\text{GNLS}}\}$  on three base dimensions give rank 3, nullity 1:

$$a = \frac{\hbar}{m_{\text{GNLS}}c_{s0}}, \quad a/\xi_h = 1/\sqrt{2}.$$

**Verdict.** **RETAIN\_L\_T\_M.** Computed from two conjuncts: the  $\{L, T\}$ -rejection gate (dropping  $M$  leaves  $\geq 1$  non-dimensionless target, needing  $M$ -bearing conversion factors  $LT^2M$ ,  $L^2T^2M$ ,  $L^2T^2$ ) and the mass fork ( $m_{\text{GNLS}}$  explicit of dimension  $M$ ;  $m_{\text{defect}}$  not derived, **INFLOW\_MASS\_SOURCE\_MISSING**). The  $\{L, T\}$  projection is a natural-unit representation only, not a base-system change.

**Carried residuals (first-class, not repaired).** **INFLOW\_MASS\_SOURCE\_MISSING** ( $m_{\text{defect}}$  emergence deferred to pathA\_21;  $\hbar J/c_\gamma^2 = M$  is dimensional-only), **LT\_R\_norm\_gate\_fails\_without\_new\_conversion\_factor** (**REJECTS\_TRUE\_LT\_BASE**) and the two 4D/3D  $R$ -norm conversion gaps (pathA\_22), plus **A\_PIN\_IS\_BRANCH\_MOMENT\_NO**, **EOS\_FROM\_GNLS\_FACTOR**, **NO\_NET\_ACCRETION\_BC\_UNDERIVED**, **M\_TO\_G\_UNIFICATION**, **SCALE\_MAP\_INPUTS**.

## A.2 Stage 005: EOS Sound Speed and Light/Sound Ratio as a Free Knob

**Part / anchor.** Part I – The medium

**Claim status.** REDUCED / CONTROLLED REDUCTION

**Purpose.** Derive the phonon sound speed from the imposed EOS, pin the three velocity scales and the  $c = c_\gamma$  wave-sector ceiling, and land the light/sound ratio  $\lambda_\gamma = c_\gamma/c_s$  as a free calibration input (**C\_GAMMA\_RATIO\_UNDERDETERMINED**).

**Status.** REDUCED / CONTROLLED REDUCTION. The sound-speed law and the  $c = c_\gamma$  ceiling are exact symbolic results *within the imposed EOS closure*; the honest headline is the **CALIBRATED** landing:  $\lambda_\gamma = c_\gamma/c_s$  is *unpinned by the parent action* – a free calibration input, with the tail  $(c/c_s)^3 = \lambda_\gamma^3$  carried symbolic (not set to 1). Consuming stage: it cites **ledger\_stage004** (I-1)’s dictionary and **EOS\_FROM\_GNLS\_FACTOR** handoff ( $h_0, \xi_h$ ) and owns the sound-speed derivation I-1 deferred here.

**Sound speed from the EOS.** From the imposed  $P = K\rho^5$  (**EOS\_CLOSURE\_IMPOSED**) and the barotropic definition, by symbolic differentiation:

$$c_s^2(\rho) = \frac{1}{m_{\text{GNLS}}} \frac{dP}{d\rho} = \frac{5K\rho^4}{m_{\text{GNLS}}}, \quad [c_s] = LT^{-1}, \quad \frac{d \ln c_s}{d \ln \rho} = 2.$$

The factor 5 falls out of the derivative. **EARNED** relative to the imposed EOS.

**Three speeds and the  $c = c_\gamma$  ceiling. Velocity scales.**  $v_b = (\hbar/m_{\text{GNLS}})\nabla\theta$ ,  $c_s$ ,  $c_\gamma$ , each of dimension  $LT^{-1}$  ( $v_b$ ’s pin consumed from I-1). **Ceiling (derived from the wave sector).** For the trapped mode  $\omega = c_\gamma\sqrt{k^2 + k_\perp^2}$ ,

$$\frac{d\omega}{dk} = \frac{c_\gamma k}{\sqrt{k^2 + k_\perp^2}}, \quad c_\gamma^2 - \left(\frac{d\omega}{dk}\right)^2 = \frac{c_\gamma^2 k_\perp^2}{k^2 + k_\perp^2} \geq 0,$$

strictly positive for  $k_\perp \neq 0$ : the group speed is bounded by  $c_\gamma$  and approaches it. The bound-mode clock  $\omega_0 = c_\gamma k_\perp$  gives the boosted internal rate  $\omega_0/\gamma$  (time dilation) with no  $E = mc^2$  premise.

**Coupled principal symbol (pathA\_20b support).** On the neutralized homogeneous background the coupled symbol block-diagonalizes: the Madelung phonon determinant  $\det M = \hbar(\omega^2 - c_{s0}^2 k^2)$  with  $c_{s0}^2 = 5K\rho_0^4/m_{\text{GNLS}}$  falling out via  $h' = 5K\rho_0^3$ , the gauge block  $C_E\omega^2 - C_B k^2 = C_E(\omega^2 - c_{\text{bulk}}^2 k^2)$  ( $c_{\text{bulk}}^2 = C_B/C_E$ ), and  $\det P = \det M \cdot P_T^2$ . The off-diagonal current/London terms are lower order and do not set the cone (BULK\_PRINCIPAL\_TRANSVERSE\_BRANCH\_ESTABLISHED);  $[c_s] = [c_\gamma]$  is labeled *non-evidentiary* for equality.

**The free-knob landing and its negative control. C\_GAMMA\_RATIO\_UNDERDETERMINED.**

Computed, not asserted. With  $C_E, C_B, K, \rho_0, m_{\text{GNLS}}$  independent, the residual  $C_B/C_E - 5K\rho_0^4/m_{\text{GNLS}}$  is not identically zero and no source equation is present, so `forced_equals_valid = False`  $\Rightarrow$  `FORCED_EQUALITY_REJECTED_WITHOUT_SOURCE_EQUATION`. It is *reversible*: inserting the identity  $C_B \rightarrow 5K\rho_0^4 C_E/m_{\text{GNLS}}$  flips it to `C_GAMMA_EQUALS_C_S` – so `EQUALS` is reachable only via an actually-inserted source equation the action does not supply. Two-layer sharpening: bulk `C_GAMMA_BULK_UNDERDETERMINED`, brane `C_GAMMA_RATIO_STILL_UNDERDETERMINED`; if  $c_{\text{bulk}}$  is  $\rho_0$ -independent then  $\lambda_\gamma \propto \rho_0^{-2}$  ( $d \ln \lambda_\gamma / d \ln \rho_0 = -2$ ).

**Carried residuals (first-class, not repaired).** `EOS_CLOSURE_IMPOSED`; `C_GAMMA_RATIO_UNDERDETERMINED`; `STATIONARY_PROFILE_UNDERDETERMINED_BY_BRANCH_DATA`; `NO_NET_ACCRETION_BC_UNDERIVED`; `HBAR_PROVENANCE_U`; `HBAR_FREE_SUBSTRATE_RELATION_MISSING`; `H_2PI_RATE_CLASSIFICATION_UNDERDETERMINED`; `BULK_METRIC_SPEED`; `PARENT_METRIC_ACOUSTIC_IDENTIFICATION_MISSING`; `BRANE_ZERO_MODE_REDUCTION_UNDERIVED`; `BRANE_PHOTON_CONE_REQUIRES_PROFILE`. The mass bridge  $\hbar J/c_\gamma^2 = M$  is a labeled dimensional candidate only, not a  $m_{\text{defect}}$  derivation.

### A.3 Stage 006: Two-Phase Material-State Ontology (Order Field $\chi_B$ )

**Part / anchor.** Part I – The medium

**Claim status.** EXACT WITHIN CLOSURE

**Purpose.** Formalize the brane/bulk ontology as a POSTULATED two-phase action – one conserved medium, order field  $\chi_B \in [0, 1]$ ; brane = ordered shear-supporting phase, bulk = the same medium de-structured, throat = phase-conversion site – dimensionally classified (`ACTION_SPECIFIED_CLASSIFIED`), with the recovery reduction to the frozen old-ledger  $S_{\text{leak}}$  verified assert-zero and the  $\theta$ -as-Maxwell- $\varphi$  no-go carried able-to-fail.

**Status.** EXACT WITHIN CLOSURE. **The wall is made explicit as a postulated field – it is NOT derived from the one medium.** Every term of the free-energy functional and order-state balance is POSTULATED and labeled (13 pinned modeling choices P1–P13); the EARNED content is exact *within that imposed closure*: structural-closure identities, single-kink wall admission, and the recovery reduction against the frozen firewall. Cross-engine agreement proves consistent encoding of the same postulated  $F$ , not derivation. Consuming stage: cites `ledger_stage004` (dictionary, single-well  $U$ ), `ledger_stage005` ( $c_{s0}^2$ ), and `ledger_stage003` ( $c_\gamma^2 = \mu_R/\rho_{br}$ ,  $C_J = -J\rho_{B0}$ ,  $B_{\text{eff}} = \rho_{B0}^2/\chi_c$ , the second-class  $\Rightarrow$  stray-DOF classification rule) with exact-value citation-integrity checks.

**The postulated action and balance.**

$$F = \int d^4X \left[ \frac{1}{2} m_{\text{GNLS}} n |\mathbf{u}|^2 + U(n) + f_B(\chi_B) + \frac{\kappa_B}{2} |\nabla_4 \chi_B|^2 + \chi_B f_{\text{shear}} + f_{\text{throat}} + f_{\text{mix}} \right],$$

$$\partial_t n + \nabla_4 \cdot (n \mathbf{u}) = 0, \quad \partial_t (\chi_B n) + \nabla_4 \cdot (\chi_B n \mathbf{u} + \mathbf{J}_\chi) = n \Gamma_B, \quad [\Gamma_B] = T^{-1}.$$

Key pins: kinetic term carries the constituent mass (KINETIC\_MASS\_FACTOR\_PINNED;  $[n] = L^{-4}$ ); double-well  $f_B = a_B \chi_B^2 (1 - \chi_B)^2$  with minima  $\{0, 1\}$ ,  $n$ -independent (SAME\_DENSITY\_DEGENERACY\_POSTULATED) – required because the cited  $U(\rho) = (K/4)\rho^5$  is *single*-well; MacCullagh-form shear gate on the displacement curl;  $\chi_B \perp P$  (POSTULATED\_ANISOTROPY dims-only);  $\mathbf{J}_\chi = 0$  default; global returns  $R_0 = -M_0$ ,  $R_1 = -D_1$  printed as postulates, not asserted locally; throat = phase-conversion ontology, not a solve. Every integrand term audited to  $ML^{-2}T^{-2}$ ; adjunct rows ( $\mu_\chi, M_\chi, P_{\text{order}}, M_n$ ) dimension-pinned, with the energy ledger fixed to the variationally consistent  $P_{\text{order}} = \int \mu_\chi D_t \chi_B d^4 X$  (HANDOFF\_P\_ORDER\_N\_PLACEMENT\_CORRECTED).

**Structural closure and wall admission (EARNED within closure).** The advective form  $D_t \chi_B = \Gamma_B - (1/n) \nabla_4 \cdot \mathbf{J}_\chi$ , the disorder complement for  $n_D = (1 - \chi_B)n$ , and the  $\Gamma_B$ -cancelling sum reproducing total conservation are derived by real derivative expansion in both engines. The static EL equation  $\kappa_B \chi_B'' = f_B'$  admits the kink  $\chi_B = \frac{1}{2}(1 + \tanh(w/2\delta))$  with solved width  $\delta = \sqrt{\kappa_B/2a_B}$  and surface tension  $\sigma_{\text{wall}} = \sqrt{2a_B \kappa_B}/6$ ,  $[\sigma_{\text{wall}}] = ML^{-1}T^{-2}$ . **Slab caveat (carried).** Bulk on both sides makes the brane a finite kink–antikink *slab*; the double-well selects no slab width –  $W_{\text{slab}}$  is an un-earned input mapping onto the old ledger’s  $L/a$  self-selection item (sim-deferred). Wall admission  $\neq$  slab stability.

**Recovery reduction (EARNED within closure; the frozen firewall).** Projection with unit-normalized  $W(w)$  and in-engine IBP gives the two-source law  $\partial_t \rho_B + \nabla_3 \cdot \mathbf{J}_B = S_{\text{flux}} + S_{\text{convert}}$  with the general  $\chi_B(w), \Gamma_B(w), J_\chi^w(w)$  computationally live; the in-engine limit  $\chi_B \rightarrow 1, \Gamma_B \rightarrow 0, \mathbf{J}_\chi \rightarrow 0$  reduces it assert-zero to the *independently constructed* frozen target  $S_{\text{leak}} = -[W j^w]_\partial + \int W' j^w dw$ ,  $j^w = n u^w$  (stage\_243), including the stage\_244 Gaussian one-mode closed form  $S_{\text{leak}} = \sqrt{2} \mu_w \rho_0 \mathcal{E}_0 / (2\sqrt{\pi} \lambda^3)$  re-derived by direct integration (RECOVERY\_REDUCTION\_VERIFIED). Conditionality computed:  $\chi_B = c \neq 1$ ,  $\Gamma_B \neq 0$ ,  $J_\chi^w \neq 0$  each leave nonzero computed residuals – the limit does real work.

**The  $\theta$ -as-Maxwell- $\varphi$  no-go (carried FATAL\_FLAW).** A stable order-parameter phase has positive gradient energy ( $\kappa_{\text{phase}} > 0$ )  $\Rightarrow K_\theta = -\kappa_{\text{phase}} < 0$  in the Lagrangian; Maxwell’s electric square needs  $K_\theta = C_J^2/\rho_{br} > 0$  – opposite signs (exact predicate). The constraint bracket  $\{\Phi_1, \Phi_2\} = k^2(J^2 \rho_{B0}^2 + \kappa_{\text{phase}} \rho_{br})/\rho_{br}$  has its zero locus solved in-engine at exactly  $K_\theta = +J^2 \rho_{B0}^2/\rho_{br}$ : via the consumed stage\_003 rule the provenance branch lands FAIL\_CAUCHY\_STRAY\_LONGITUDINAL (finite  $B_{\text{eff}} = \rho_{B0}^2/\chi_c$ ), the  $B_{\text{eff}} = 0$  branch FAIL\_C5\_LONGITUDINAL\_ZERO\_MODE, and the tuned fixture ( $K_\theta = J^2 \rho_{B0}^2/\rho_{br}, B_{\text{eff}} = 0, m_\theta^2 = 0$ ) C5\_RESOLVED\_MAXWELL\_BY\_TUNING – flagged BY\_TUNING, never WITH\_PROVENANCE (flag computed from provenance booleans; detunings re-fire the pathA\_36 control tokens). The only provenance sign-flip ( $\kappa_{\text{phase}} < 0$ ) is a Lifshitz instability,  $k_*^2 = -\kappa_{\text{phase}}/2\kappa_4 > 0 \iff \kappa_{\text{phase}} < 0$  – the already-killed pathA\_25 wall (FAIL\_LIGHT\_STARVED lineage). The transverse sector is untouched: the  $\varepsilon$ -parametrized dispersion derived in-engine gives  $\omega^2 = (\mu_R/\varepsilon)k^2$ ;  $\varepsilon = \rho_{br}$  reproduces the consumed  $c_\gamma^2$  (PASS\_TRANSVERSE\_UNDISTURBED),  $\varepsilon \neq \rho_{br}$  fires FAIL\_TRANSVERSE\_DISTURBED. **This stage does not earn light** – the no-go is the point.

**Drift and carried items (first-class, not repaired).** DRIFT(6) computed:  $\{\chi_B; a_B; \kappa_B; \alpha_{\text{aniso}}; \Gamma_B$ ; gating struct (rung-W reconciliation printed; THETA\_BRANCH\_DEAD\_NOT\_ADMITTED;  $\rho_{B0}, \chi_c$  cross-referenced to the pathA\_41 Part-VI drift trio, not double-counted). Carried: SECOND\_MEDIUM\_DRIFT lineage, the slab width  $W_{\text{slab}}$ , Gate-L  $\delta w = u_w$  translation Goldstone (deferred),  $\mathbf{J}_\chi \neq 0$ ,  $f_{\text{throat}}/f_{\text{mix}}$  placeholders, dynamics/energy-ledger adjunct. Two source-prose dimensional defects pinned and ablation-documented (kinetic mass factor;  $P_{\text{order}}$   $n$ -placement).

## A.4 Stage 007: Shear-Surface G0 Freeze (Frozen Medium Action, Drift Ledger, DOF)

**Part / anchor.** Part I – The medium (closing Part-I stage)

**Claim status.** EXACT WITHIN CLOSURE

**Purpose.** Formal home of the pathA\_35 G0 shear-surface *constitutive* freeze: the frozen brane/light action ( $L_{\text{Mac}} + L_{Pu} + L_{uw}$  under the Gaussian profile  $g_\ell(w)$ ), hash-guarded (T0\_SHEAR\_FROZEN(d9520d3819c3f7...), with the honest drift ledger SECOND\_MEDIUM\_DRIFT\_AT\_FREEZE(11) computed from an enumerated table and the flat-brane linear DOF = 8 rank-computed.

**Status.** EXACT WITHIN CLOSURE. **The 11 freeze inputs are POSTULATED/CALIBRATED – the freeze freezes *terms*, not gate answers.** The EARNED content is exact within that declared closure: (i) *freeze fidelity* – both engines re-extract the canonical **freeze-action** blocks by byte-exact fence-parsing (never fixed offsets) and assert SHA-256 equality (G0 d9520d3819c3f7..., T0 8fa41ac51e88... byte-embedded); (ii) the *flat-brane linear DOF* = 8 computed from rank/nullity bookkeeping (in-plane  $u^a$  curl-transverse 2 + kinetic-curl 1; T0  $P^i$  tangent 3 + radial 1;  $u_w$  1; C5  $\varphi$  0 – the absent longitudinal partner the later C5 crux attacks); (iii) the *dimensional firewall* over every frozen term. This stage computes **no** gate verdict (Gate-L excluded; exposure names carried as prose provenance only) and does **not** earn light (Stage 003 derives the two transverse photons at  $c_\gamma^2 = \mu_R/\rho_{br}$  from this stage's frozen  $L_{\text{Mac}}$ ).

**The frozen action.** Kept (0 new): the GNLS parent action and the T0 polar-OP action (byte-for-byte, hash-embedded). Frozen brane blocks under  $S_{\text{brane}} = \int dt d^4X g_\ell(w) [L_{\text{Mac}} + L_{Pu} + L_{uw}]$ :

$$L_{\text{Mac}} = \frac{1}{2}\rho_{br}(\partial_t u^a)^2 - \frac{1}{2}\mu_R \Omega_u^a \Omega_u^a, \quad L_{Pu} = -\lambda_{Pu} \varpi_a \Omega_u^a, \quad \varpi_a := (\hat{w} \times P_\parallel)_a,$$

$$L_{uw} = \frac{1}{2}\rho_{br}(\partial_t u_w)^2 - \frac{1}{2}\rho_{br}\Omega_w^2 u_w^2, \quad g_\ell(w) = \frac{e^{-(w/\ell_g)^2}}{\sqrt{\pi}\ell_g}, \quad \int g_\ell dw = 1 \text{ (derived)}.$$

Preserved verbatim:  $L_{Pu}$  *re-admits the  $\epsilon$ -contracted/chiral class excluded by T0* and requires the conceded axis  $\hat{w}$  – a structural-postulate cost, not a free choice.

**The drift ledger (computed).** Both engines enumerate the 11 and *compute* the subcounts and the verdict string (the source gate's hardcoded literals are retired): **4 constants**  $\rho_{br} [ML^{-3}]$ ,  $\mu_R [ML^{-1}T^{-2}]$ ,  $\lambda_{Pu} [ML^{-1}T^{-2}]$ ,  $\Omega_w [T^{-1}]$  (table dims asserted against the firewall targets); **1 function**  $g_\ell(w; \ell_g)$  (fixed Gaussian, one width, target-blind admission); **6 structural postulates** (conceded  $\hat{w}$ +wall; free-slip  $u^a$ ;  $P^i$  reused as Cosserat reservoir; massless spin-wave baseline; the parity-even  $P$ - $u$  operator; no C5  $\varphi$ ). New-field content ( $u^a$  3 +  $u_w$  1 = 4 DOF) is counted *separately*. **Erratum (2026-07-04), carried first-class: the 11 STANDS** – no  $\rho_{br}$  overcount (pathA\_25's  $\varrho_{br}[\rho]$  belongs to the closed density-smectic candidate, OUT\_OF\_ACTIVE\_NG5);  $\{\rho_{br}, \mu_R\}$  carry the registered-pending Route-A reduction (ROUTE\_A\_UNDERDETERMINED\_MISSING\_NONLINEAR\_THROAT, register R10); the honest cross-sector drift  $\{\rho_{B0}, \chi_c, C_{hu}\}$  is a Part-VI (pathA\_41) item, guarded in-engine against absorption into the 11.

**Supersession and firewalls.** Both facts asserted: the Stage 006  $\chi_B$  wall superseded the fixed-shape  $g_\ell(w)$  as the *material-state* closure, while G0 *remains* the light-sector *constitutive* freeze (Stage 003 consumes  $L_{\text{Mac}}$  as-is) – a scope split (register R21), not a reduction. Notational firewall (register R22):  $\mu_R$  (3D brane,  $ML^{-1}T^{-2}$ ) and Stage 006’s  $\mu_R^{(4)}$  (4D density,  $ML^{-2}T^{-2}$ ) are asserted dimensionally distinct, related only by the *pending* R17 projection  $\mu_R = \int \chi_B \mu_R^{(4)} dw$  (dim-consistency asserted).

**Verification.** Dual-engine, independent routes (the source pair’s `-compare` JSON payload mirror is retired): SymPy 96 PASS / Mathematica 94 PASS, both exit 0, CWD-independent. Able-to-fail teeth all fire: the five source-gate ablations (2 dimensional, 3 DOF each recomputing  $7 \neq 8$ ), two hash teeth (in-block byte corruption; missing fence), five enumeration teeth (drop entry; miscategorize; inject  $\rho_{B0}$ ; corrupt  $n$  pre-verdict; corrupt table dim), and the  $\mu_R$  firewall tooth. Full tri-review on fresh agents: arbiter re-run, `FIDELITY_CLEAN` (block-by-block coverage diff, no dropped check), `ADVERSARIAL_CLEAN` (26-run mutation matrix: hash genuinely fence-parsed – one byte inside the block fails, one byte outside passes; dual-corruption of both engines fails both; no cross-engine comparison anywhere).

## Appendix B

# Part II Stage Cards

### B.1 Stage 001: Solid-Angle and Second-Moment Primitives

**Part / anchor.** Part II – Gravity

**Claim status.** EXACT

**Purpose.** Derive the unit-sphere solid angles and normalized second moments used by the pathA\_21c matter-stress force assembly.

**Status.** EXACT. This stage is closed geometry: no profile, normalization, or field-equation residual remains inside the stated claims.

**Derivation.** For  $S^2 \subset \mathbb{R}^3$ , take

$$n(\theta, \phi) = (\cos \theta, \sin \theta \cos \phi, \sin \theta \sin \phi), \quad dS = \sin \theta \, d\phi \, d\theta.$$

Then **Omega2 solid angle**.

$$\Omega_2 = \int_0^\pi \int_0^{2\pi} \sin \theta \, d\phi \, d\theta = 4\pi.$$

With  $n_1 = \cos \theta$  and  $n_2 = \sin \theta \cos \phi$ , **S2 second moments**.

$$\langle n_1^2 \rangle_{S^2} = \frac{1}{4\pi} \int_0^\pi \int_0^{2\pi} \cos^2 \theta \sin \theta \, d\phi \, d\theta = \frac{1}{3}, \quad \langle n_1 n_2 \rangle_{S^2} = 0.$$

The cross moment vanishes because  $\int_0^{2\pi} \cos \phi \, d\phi = 0$ .

For  $S^3 \subset \mathbb{R}^4$ , take

$$n(\chi, \theta, \phi) = (\cos \chi, \sin \chi \cos \theta, \sin \chi \sin \theta \cos \phi, \sin \chi \sin \theta \sin \phi), \quad dS = \sin^2 \chi \sin \theta \, d\phi \, d\theta \, d\chi.$$

Then **Omega3 solid angle**.

$$\Omega_3 = \int_0^\pi \int_0^\pi \int_0^{2\pi} \sin^2 \chi \sin \theta \, d\phi \, d\theta \, d\chi = 2\pi^2.$$

With  $n_1 = \cos \chi$  and  $n_2 = \sin \chi \cos \theta$ , **S3 second moments**.

$$\langle n_1^2 \rangle_{S^3} = \frac{1}{2\pi^2} \int_0^\pi \int_0^\pi \int_0^{2\pi} \cos^2 \chi \sin^2 \chi \sin \theta \, d\phi \, d\theta \, d\chi = \frac{1}{4}, \quad \langle n_1 n_2 \rangle_{S^3} = 0.$$

The cross moment vanishes because  $\int_0^\pi \cos \theta \sin \theta \, d\theta = 0$ .

**Physical use.**  $\Omega_2 = 4\pi$  and  $\Omega_3 = 2\pi^2$  set the reduced  $r^{-2}$  and bulk  $R^{-3}$  Gauss normalizations. The second moments give the  $\delta_{ij}/d$  angular average used in the convective factor  $-(1 + 1/d)$  and the pressure factor  $1/d$ .

## B.2 Stage 002: Matter-Stress Inter-Defect Force Assembly

**Part / anchor.** Part II – Gravity

**Claim status.** REDUCED / CONTROLLED REDUCTION

**Purpose.** Assemble the inter-defect force between two drain defects from the matter (Noether) stress, earning the bilinear  $Q_1 Q_2$  structure, the reduced-3D  $r^{-2}$  and bulk-4D  $R^{-3}$  power laws, and the attractive like-drain sign, consuming the Stage 001 geometry primitives.

**Status.** REDUCED / CONTROLLED REDUCTION. Three verdict layers are carried from the earned source: the leading matter-stress sign `FORCE_ATTRACTIVE_DERIVED` (target-blind); the full sign `SIGN_RESIDUAL_QUANTUM_VCONF_MAXWELL_PROFILE` (honest residual); acceptance `PASS_WITH_NAMED_RESIDUALS`. The bilinear structure, the power laws, and the attractive sign are earned (form + sign); the overall magnitude is a calibration knob; the full sign carries named, profile/sim-deferred residuals.

**Derivation.** The medium obeys  $P = K\rho^5$ ,  $h = (5K/4)\rho^4$ , with the barotropic identity  $dP/d\rho = \rho(dh/d\rho)$ . The matter (Noether) stress in Madelung form is

$$\Pi_{ij} = m\rho v_i v_j + \delta_{ij} P(\rho) + \sigma_{ij}^Q, \quad \sigma_{ij}^Q = \frac{\hbar^2}{4m} \left[ \frac{(\partial_i \rho)(\partial_j \rho)}{\rho} - \partial_i \partial_j \rho \right],$$

with the quantum-stress divergence contributing no net far-field force at this order. A defect draining flux  $Q$  sets up, by Gauss's law, the inflow

$$v = -\frac{Q \hat{n}}{\Omega_{d-1} r^{d-1}},$$

where  $\Omega_2 = 4\pi$  ( $d = 3$ ) and  $\Omega_3 = 2\pi^2$  ( $d = 4$ ) are the Stage 001 solid angles. The Bernoulli cross-pressure of the two overlapping inflows is  $\delta P = -m\rho(v_1 \cdot v_2)$ . Integrating the traction  $\Pi_{ij} n_j$  over a control sphere around defect 2 and using the Stage 001 second moment  $\langle n_i n_j \rangle = \delta_{ij}/d$  gives the convective factor  $-(1 + 1/d)$  and the pressure factor  $+1/d$ , whose sum is the total flux  $-mNQ_2 v_1$  (the  $1/d$  pieces cancel). The stationary force  $F_{12} = -\oint \Pi_{ij} n_j dS$ , with defect 1's own inflow substituted, yields the two lanes.

**Reduced-3D force law.**

$$F_{12} = -\frac{m N_3 Q_1 Q_2}{4\pi r_{12}^2} \hat{r}_{12}.$$

**Bulk-4D force law.**

$$F_{12} = -\frac{m \rho_4 Q_1 Q_2}{2\pi^2 R_{12}^3} \hat{R}_{12}.$$

**Attractive sign.** Since  $F \propto -Q_1 Q_2$ , like drains ( $Q_1 Q_2 > 0$ ) attract and opposite signs repel; the sign is derived from the Gauss-drain / Bernoulli / surface-integral chain, not assumed.



**Physical use and named residuals.** Target-blind, the inter-defect force is bilinear in the drain strengths, falls as  $r^{-2}$  (reduced 3D) and  $R^{-3}$  (bulk 4D), and is attractive between like drains. Honest scope: the overall normalization ( $I_F/\Theta_Q/\text{branch-profile}$ ) is a *calibration* knob, not derived; and the full sign carries the QUANTUM\_STRESS\_FAR\_FIELD, VCONF\_BODY\_FORCE, and MAXWELL\_Z\_GAUGE\_JEXT residuals, which are profile/sim-deferred and first-class.

### B.3 Stage 008: Monopole/Dipole Return Constraint Specification

**Part / anchor.** Part II – Gravity (first return/radiation stage)

**Claim status.** OPEN

**Purpose.** Render the old ledger’s open-item #9 as an exact constraint specification: the raw  $\ell = 0/1/2$  outgoing  $c_s$ -wave amplitudes and radiation orders, and the precise source moments any brane $\leftrightarrow$ bulk return must cancel (MONOPOLE\_DIPOLE\_RETURN\_CONDITIONAL). The formal home of the targets  $R_0 = -M_0$ ,  $R_{1,i} = -D_{1,i}$  consumed by Stages 009/010 (pathA\_29) and 022/023 (pathA\_34); register edge R23.

**Status.** OPEN – deliberately. **This is a verified CONSTRAINT-SPECIFICATION, not a falsifiable test of no-monopole-radiation** (the source gate’s own scope caveat, carried verbatim-class). GR forbids monopole/dipole gravitational radiation; the drain breaks brane mass conservation, so suppression is conditional on a return that cancels the moments below. The cancellation condition is algebraic bookkeeping (an  $x - x$  identity, labeled as such in both engines); **cancellation\_possible** is a literal parameter flag printed as *SCOPE: parameter, not computed – track-3 decides*. The falsification lives downstream (Stages 009/010; the Gate-6  $Z_{0,\text{ret}}/Z_{1,\text{ret}}$  selector).

**The DtN ladder (earned; scanned, not typed).** From spherical Hankel closed forms  $h_\ell = j_\ell + iy_\ell$ , both engines form  $\Lambda_\ell = k h'_\ell/h_\ell|_{z=ka}$ , normalize the admittance by its static value, and *scan* for the first purely-imaginary  $k$ -coefficient:

$$p_{\text{raw}} = 1, 3, 5; \quad \text{ kernels } i\frac{\omega}{c_S}a, \quad i\frac{\omega^3}{c_S^3}\frac{a^3}{2}, \quad i\frac{\omega^5}{c_S^5}\frac{a^5}{27},$$

$$A_{\text{raw}}^{(\ell 0)} = \frac{iM_0a\omega}{c_S}, \quad A_{\text{raw}}^{(\ell 1)} = \frac{iD_1a^3\omega^3}{2c_S^3}, \quad A_{\text{raw}}^{(\ell 2)} = \frac{iQ_2a^5\omega^5}{27c_S^5},$$

all symbolically nonzero; steady control  $\lim_{\omega \rightarrow 0} A^{(\ell 0)} = 0$ ; dominance  $p(\ell=0) < p(\ell=2)$  from the scanned powers.

**Cancellation targets (bookkeeping, labeled).**  $R_0(\omega) = -M_0(\omega)$  with  $M_0 = \int_{\text{brane}} S_{\text{leak}} d^3x$  (cancels  $O(\omega^1)$ );  $R_{1,i}(\omega) = -D_{1,i}(\omega)$  with  $D_{1,i} = \int x_i S_{\text{leak}} d^3x + \int j_i d^3x$  including the carried odd wake (cancels  $O(\omega^3)$ ). Conditions, not closure. The derivative outlet vertex  $g_{W0} = \eta\omega$  is recorded as a **branch\_assumption**, not verdict-bearing. Live-source consistency is anchored by direct integration of the frozen Stage-243/244 closed forms ( $\int W' j^w dw = -\sqrt{2} \ell_w j_0/4$ ;  $S_{\text{leak}} = \sqrt{2} \mu_w \rho_0 E_0/(2\sqrt{\pi} \lambda^3)$ ) with the strict-recovery limits computed but *not* taken as a basis; Stage 006 owns the recovery-reduction derivation (boundary printed).

**Verdict machinery and guards.** The verdict is computed from computed booleans plus the literal flag; the synthetic control (True, False, False)  $\rightarrow$  MONOPOLE\_RADIATION\_UNAVOIDABLE proves the rung reachable. The nine source controls are carried honestly: four computed, five printed as declarations (not counted checks), with the source’s framing – guards against obvious tautologies, not evidence of suppression-vs-unavoidable. Consumed from Stage 005:  $c_s^2 = 5K\rho^4/m_{\text{GNLS}}$  (cited, exact-value integrity check  $5 \cdot \frac{1}{500} \cdot (\sqrt{10})^4 = 1$ ); frozen slice  $G = c = c_s = 1$ ,  $K_{\text{eos}} = 1/500$ ,  $(a^*, L^*)$  exact rationals, cited.  $Q_2$  is exported as a **free anchor** (not derived  $\ell = 2$  physics) for Stage 026.

**Verification.** Dual-engine, independent routes, zero file I/O (the source pair’s JSON/YAML engine bridge is retired): SymPy 54 PASS / Mathematica 57 PASS (3 = arity self-checks), both exit 0, CWD-independent. Eight able-to-fail teeth fire (Hankel-form corruption trips the scan; kernel-coefficient, moment-kill, bookkeeping-integrity, verdict-machinery, anchor-prefactor, steady-limit, consumed-law corruptions). Full tri-review on fresh agents: FIDELITY\_CLEAN (coverage-diff, math re-derived by hand) and ADVERSARIAL\_CLEAN (14/14 mutation matrix; dual-corruption fails both engines; arity scan incl. planted-mismatch detection), followed by a nit remediation (tally hygiene, restored recovery-slice observations, pipeline-routed tooth) re-verified clean.

## B.4 Stage 009: Flat-Slab Return Residual (Check A)

**Part / anchor.** Part II – Gravity (brane $\leftrightarrow$ bulk return, Check A)

**Claim status.** EXACT WITHIN CLOSURE

**Purpose.** Derive, on the postulated flat-slab family, what a finite DC-sink brane $\leftrightarrow$ bulk return actually delivers against the Stage-008 cancellation targets: the return transfer  $T_\ell(\omega) = \alpha_\ell e^{i\omega 2d/c_s}$ , the bounded residual orders  $p_{\text{res}}(\ell=0) = 1$ ,  $p_{\text{res}}(\ell=1) = 3$ , and the signed local source accounting  $Z = -M_0\varepsilon_0/(1 + \varepsilon_0)$  – the Check-A component of RETURN\_RESIDUAL\_PREDICTION (Check-B localization = Stage 010).

**Status.** EXACT WITHIN CLOSURE – exact *within the postulated slab family* (brane at  $w = 0$ , return/absorber at  $w = d$ ; the geometry is postulated, the return response is derived on it). The v3 labeling is carried exactly:  $Z < 0$  (drain admissibility) is the **premise** (Z\_is\_premise=true); the formula  $Z = -M_0\varepsilon_0/(1 + \varepsilon_0)$  is **accounting** (channel sum  $Z_{\text{throat}} + Z_{\text{return}}$ , sign certificate computed). Open-item #9 is *sharpened, not closed*; the headline is printed only as the Check-A component with the Stage-010 qualifier.

**The derived return response.** From the bidirectional Helmholtz basis  $\Phi_\ell = A_\ell e^{i\omega w/c_s} + B_\ell e^{-i\omega w/c_s}$ , the round-trip transit time is *solved* ( $\tau = 2d/c_s$ , asserted via  $e^{i\omega\tau} \equiv \text{phase}$ ). DC continuity fractions  $\alpha_\ell = 1/(1 + \varepsilon_\ell)$  with independent per-channel transmissions  $\varepsilon_0, \varepsilon_1 > 0$ ; steady circulation balances ( $J_{\text{leak}} = J_{\text{return}} + J_{\text{neighbor}}$ , both moments) assert-zero. A genuine  $\omega$ -order scan computes  $\nu_\ell = \text{ord}_\omega(1 - T_\ell) = 0$  – a finite DC sink leaves the  $\omega^0$  deviation  $1 - T_\ell(0) = \varepsilon_\ell/(1 + \varepsilon_\ell)$  – so the residual amplitudes  $A_{\text{res}} = \text{kernel}_\ell \cdot \text{moment} \cdot (1 - T_\ell)$  sit at **computed** orders  $p_{\text{res}} = p_{\text{raw}} + \nu_\ell = 1, 3$ : a *bounded monopole/dipole  $c_s$ -radiation residual tied to the drain strength* – the falsifiable departure from GR. The Check-A classification booleans are computed from the orders.

**Reversibility (per-channel, computed).** At  $\varepsilon_0 \rightarrow 0^+$ :  $T_0 \rightarrow e^{i\omega\tau}$ , strict  $\nu_0 = 1 \Rightarrow p_{\text{res},0} = 2$ , and  $Z \rightarrow 0$  (a perfect-return channel has no drain). At  $\varepsilon_1 \rightarrow 0^+$ : strict  $\nu_1 = 1$  computed *directly from the  $\ell = 1$  limit* (the source reused  $\nu_0$ ; the reshape computes it independently)  $\Rightarrow p_{\text{res},1} = 4$ . The

prediction is contingent on the transmissions, not baked in. Consumed from Stage 008 (cited, dual-site integrity-checked): the three radiation kernels;  $Q_2/\text{kernel}_2$  inert (**T2\_applied=false**). Register: rows  $\{d, \varepsilon_0, \varepsilon_1\}$  (slab-family parameters;  $\varepsilon$ 's FREE-UNREDUCED on the Gate-6 route) + edge R24.

**Verification.** Dual-engine, independent routes, zero file I/O – the source pair's JSON-digest bridge, its runtime YAML reads of pathA\_28 outputs, and its SHA-256 trace bookkeeping are all retired. SymPy 48 PASS / Mathematica 52 PASS, exit 0, CWD-independent. Eight able-to-fail teeth fire. Full tri-review on fresh agents: FIDELITY\_CLEAN (all values hand-re-derived; 009/010 boundary clean); the adversarial leg (16-mutant matrix) **caught a genuine rig-class defect** – the consumed-kernel citation-integrity block was set-then-compare-to-self (a single-source corruption passed) – remediated to dual-site integrity (acceptance: all four one-site corruptions now fail in both engines) plus tally-hygiene nits, then re-verified clean. The `Off[Limit::alimv]` suppression was unmasked and shown benign (every silenced limit's result independently asserted).

## B.5 Stage 010: Slab Localization $p = 2$ with Reachable NOGO (Check B)

**Part / anchor.** Part II – Gravity (brane $\leftrightarrow$ bulk return, Check B; completes the pathA\_29 fold)

**Claim status.** EXACT WITHIN CLOSURE

**Purpose.** Show, on the postulated flat-slab family, that Newtonian  $1/r^2$  gravity SURVIVES the finite slab: both admissible DC-sink completions carry a normalizable  $m=0$  transverse zero mode whose 3D-radial Green function gives flow  $\propto 1/r^2$  ( $p = 2$ ); the anti-localizing warp control gives  $p = 3$  and the same classifier returns RETURN\_NOGO (the able-to-fail companion). This computes Check B and emits the COMPLETED joint verdict RETURN\_RESIDUAL\_PREDICTION = (Check A cited, Stage 009)  $\wedge$  (Check B computed here).

**Status.** EXACT WITHIN CLOSURE – exact *within the postulated slab family* (brane at  $w = 0$ , return/absorber at  $w = d$ ; the geometry is postulated, localization is derived on it). The gravity-range ( $1/r^2$ ) leg passes inside the localizing flat-slab family because both DC-sink completions give  $p = 2$ ; the joint verdict RETURN\_RESIDUAL\_PREDICTION is completed here, with RETURN\_NOGO genuinely reachable. Open-item #9 is *sharpened, not closed*: localization holding for the FAMILY is not the family being SELECTED by dynamics (R19/ $W_{\text{slab}}$ ); the return-cancellation targets (R23) remain deferred.

**The localizing zero mode and  $p = 2$ .** On  $0 \leq w \leq d$  each admissible completion is a transverse eigenproblem  $f'' + m^2 f = 0$  with a normalizable constant  $m=0$  zero mode ( $f_0 = 1/\sqrt{d}$ ,  $\int_0^d f_0^2 dw = 1$ ): the *destructuring/absorbing* (compact cell, Neumann) and *Bloch* ( $q=0$  periodic) cells, with gapped first modes at  $m_1 = \pi/d$ ,  $2\pi/d$ . The 3D-radial zero-mode equation  $g'' + (2/r)g' + ((\omega/c_s)^2 - m^2)g = 0$  is solved by genuine `dsolve` (dynamic route: solved first, outgoing spherical branch selected via  $C_1/C_2$  – a boundary SELECTION, normalization = compact overlap  $1/d$  – then  $\omega \rightarrow 0$ ); the limit Green  $1/(4\pi dr)$  gives flow  $1/(4\pi dr^2)$  and *extracted* exponent  $p = 2$ . The static route ( $\omega = 0$  first) gives the same  $1/(4\pi dr)$  and  $p = 2$ ; the two routes agree on both the exponent and the Green ( $\text{dynamic}_{\omega \rightarrow 0} = \text{static}$ ). The massive/gapped solve gives a Yukawa  $e^{-m_1 r}/r$  – only the  $m=0$  zero mode sets the far field. Both completions agree  $p_{\text{abs}} = p_{\text{bloch}} = 2$ .

**Able-to-fail: the counterfactual and the NOGO warp.** Multiplying the solved Green by  $r^{-4}$  ( $\rightarrow 1/r^5$ ) yields radial-operator residual  $5/(\pi dr^7) \neq 0$  – REJECTED, proving the solve has teeth. The mandatory anti-localizing warp  $\mu(w) = e^{2k_{\text{warp}}w}$  has a *non-normalizable* zero mode ( $\int_0^\infty \rightarrow \infty$ , computed) and a continuum Green  $1/(4\pi r^2)$  giving  $p_{\text{delocalizing}} = 3$ ; the same classifier returns RETURN\_NOGO. The classifier verdict is a function of the computed exponents (and the cited `quadrupole_survives`), never a stamp; RETURN\_NOGO is reachable if the return delocalizes OR the quadrupole channel does not survive.

**Consumed (cited, dual-site integrity + exact anchor).** From Stage 008:  $p_{\text{raw}}(\ell=2) = 5 \Rightarrow \text{quadrupole\_survives}$  (finite quadrupole order;  $T_2$  not applied, `T2_applied=false`, no  $\ell=2$  recompute). From Stage 009: the Check-A component `A_residual_pass=true` (joint-composition print only). Register: *zero new counted knobs*;  $k_{\text{warp}}$  is a control-construction symbol (tracked, not counted); edge R25 (localization  $p = 2$  EARNED-within-family, able-to-fail via the warp NOGO – does not discharge R19 or R23).

**Verification.** Dual-engine, independent routes, zero file I/O – the source pair’s JSON-digest bridge and all SHA-256 trace/`structure_id/expr_digest` bookkeeping are retired; the `.w1` solves the radial ODEs, the normalizations, the warp divergence, and the classifier natively (own `DSolve` basis, own `Limit/Integrate`), with an arity self-check and no silenced messages. SymPy 71 PASS / Mathematica 81 PASS (81 = 71 shared + 10 arity self-checks), exit 0, CWD-independent. Eight able-to-fail teeth fire. Full tri-review on fresh agents: `FIDELITY_CLEAN` (all values hand-re-derived; 009/010 boundary clean – Check-A cited not recomputed) and `ADVERSARIAL_CLEAN` (22-mutant matrix; a hardcoded-wrong-Green mutant confirmed the counterfactual guard is genuine, and the classifier genuinely reaches RETURN\_NOGO). The adversarial leg’s one substantive find – a *coordinated both-sites* corruption of the Stage-008 citation to another finite integer escaped – was folded to Stage-009 parity by an exact-value anchor, and re-verified (`REVERIFY_CLEAN`) as the sole gate closing that escape.

## B.6 Stage 011: Frozen-Reduction Certificate ( $L_s$ Helmholtz) (Check II-G1a)

**Part / anchor.** Part II – Gravity (brane $\leftrightarrow$ bulk return; the frozen-throat longitudinal reduction; opens the pathA\_30 D/N fold, completed by Stage 012)

**Claim status.** REDUCED / CONTROLLED REDUCTION

**Purpose.** Certify that, on the postulated straight finite throat with the wall FROZEN ( $R = R_0$ ,  $\eta = 0$ ) and the matter perturbation left dynamic ( $\propto e^{-i\omega t}$ ), the reduced longitudinal operator collapses to the constant-coefficient Helmholtz resonator operator  $L_s\psi = \psi''(s) + (\omega/c_S)^2\psi(s) = 0$ , with  $c_S^2 = 5K\rho_*^4/m$  on  $s \in [0, L_0]$  (cap  $R_0(L_0) = 0$ ) — *produced* by a per-term reduction (projection measure plus every intruding coefficient computed with its vanishing/deferral condition), NOT typed — together with the reduction certificate: the validity window under which the collapse is exact.

**Status.** REDUCED / CONTROLLED REDUCTION — a controlled reduction with an explicit validity certificate. This stage carries the reduction-certificate component of the joint pathA\_30 verdict `DN_UNITTEST_BC_DEPENDENT`; the D/N determinant, the DtN  $-(\omega/c_S)\tan(L_0\omega/c_S)$ , the half-shifted pole ladder, the Robin counterfactual, and the `BC_DEPENDENT` landing (`bc_derivation_emitted = false`, a banked calibration input) are Stage 012’s; the cut is drawn at the general-solution `dsolve` (never

called here). The wave speed is *banked* (Part I edge R1, Stage 005, at the straight-reference density  $\rho_*$ ), so this stage adds NO reduction of its own: its contributions are the two *structural* edges R26 (the validity record) and R27 (the  $\xi \neq \ell_c$  firewall), which discharge no knob.

**The produced operator (the de-rig).** The reduced operator is *assembled* from dimensionally-consistent terms,

$$L_s = \psi'' + M\psi' + N\psi - B \frac{\hbar^2}{4m^2 c_S^2} \psi'''' ,$$

with  $M = \frac{d}{ds} \log \sqrt{\gamma_0}$  ( $[M] = L^{-1}$ ), the  $\psi$ -coefficient  $N = (\omega/c_s(s))^2$  with  $c_s(s)^2 = 5K\rho_0(s)^4/m$  ( $[N] = L^{-2}$ ), and  $B \in \{0, 1\}$  the Bogoliubov inclusion flag — the  $k^4$  correction being a genuine *fourth-derivative* intrusion (from the identity  $\psi'' + (\omega/c_S)^2\psi - (\hbar^2/4m^2 c_S^2)\psi'''' = 0$  for the full BdG dispersion), NOT a coefficient shift. The three source checks are replaced by genuine computations that can each fail: `operator_is_helmholtz` is `expr_equal(L_s - ideal, 0)` on the *assembled*  $L_s$  (not two hand-typed twins); `speed_is_cs` is *extracted* from the  $\psi$ -coefficient of  $L_s$  ( $= (\omega/c_S)^2$ , bulk, not wall/healing-renormalized, not literal `True`); `domain_is_L0` is *solved* from the pinch-off  $R_0(s) = 0$  on a POSTULATED monotone taper  $R_0(s) = R_{\text{mouth}}(1 - s/L_0)$  (root =  $L_0$ ,  $R_{\text{mouth}}$ -independent; a labeled SELECTION, not `L0==L0`); and `unsuppressed_operator_intrusion` is *computed* from the assembly (not hardwired `False`) and gates the verdict.

**The reduction certificate (validity window).** Under the frozen background ( $\rho_0(s) = \rho_*$ ,  $\sqrt{\gamma_0} = A_{\perp 0}$  constant,  $A_{M0} = 0$ ) every intruding term is computed to vanish or is deferred: the projection-measure coefficient  $M = 0$ ;  $\rho'_0 = 0 \Rightarrow c'_s = 0 \Rightarrow N = (\omega/c_S)^2$  (the bulk sound speed);  $\delta V_{\text{conf}} = 0$ ; the background quantum potential  $Q(\rho_0) = -\frac{\hbar^2}{2m} \partial_s^2 \sqrt{\rho_0}/\sqrt{\rho_0}$  gives  $\partial_s Q = 0$ ; and the BdG term is deferred ( $B = 0$ ) only under  $k\xi \ll 1$  ( $\xi = \hbar/(mc_s)$ ), with smallness witness  $\hbar^2 k^2/(4m^2 c_S^2) = (k\xi/2)^2$ . Thus  $L_s$  is constant-coefficient Helmholtz *conditional on*  $\{\rho'_0/\rho_0 = 0, \sqrt{\gamma_0} \text{ const}, \delta V_{\text{conf}} = 0, \nabla Q = 0, k\xi \ll 1\}$  (edge R26); the BdG  $k^4$  term is DEFERRED, not dropped unconditionally. The healing length  $\xi = \hbar/(mc_s)$  and the confinement length  $\ell_c$  (in  $V_{\text{wall}}(\Sigma/\ell_c)$ ) are kept as DISTINCT symbols (edge R27, the firewall analogous to R22's  $\mu_R \neq \mu_R^{(4)}$ );  $\ell_c$  is INERT here ( $\delta V_{\text{conf}} = 0$ ).

**Able-to-fail teeth.** Eight teeth, mutations on copies: a nonconstant  $\sqrt{\gamma_0}$  makes  $M \neq 0$ ; a retained BdG ( $B = 1$ ) injects the  $\psi''''$  term; a nonuniform  $\rho_0(s)$  makes  $N$   $s$ -dependent (each  $\rightarrow$  `FAIL_OPERATOR_INTRUSION`); a wall/healing renormalization  $\rightarrow$  `FAIL_WRONG_SPEED`; a corrupted taper root  $\rightarrow$  `FAIL_WRONG_DOMAIN`; an  $\ell_c \rightarrow \xi$  conflation trips the firewall; a one-site R1 exponent corruption trips the dual-site integrity (a coordinated both-site drift is caught by the explicit frozen-export anchor); and a corrupt- $[K]$  mutation flips  $[c_S^2]$  to  $(3, 0, -2) \rightarrow$  `DN_UNITTEST_FAIL_DIMENSIONAL`. A dedicated witness-path tooth (nonzero  $\delta V_{\text{conf}}$  with  $M = 0$ ,  $N = \text{bulk}$ ,  $B = 0$ ) drives the verdict to `FAIL_OPERATOR_INTRUSION` through the intrusion flag alone, making the flag independently load-bearing.

**Consumed + dimensional leg.** The speed law  $c_s^2 = 5K\rho^4/m_{\text{GNLS}}$  is CITED (Part I edge R1, Stage 005) evaluated at  $\rho_*$ , integrity-checked dual-site (literal `5*K*rho**4/m` vs. the EOS route  $\partial_\rho(K\rho^5)/m$ ) plus an explicit frozen-export anchor; the EOS exponent-5 closure  $P = K\rho^5$  stays IMPOSED. The  $c_S^2$  dimensional leg uses the  $[K] = [P] - 5[\rho]$  chain in four spatial dimensions:  $[K] = (18, 1, -2)$ ,  $[c_S^2] = (2, 0, -2)$  (order  $L, M, T$ ); the corrupt- $[K]$  probe flips it to  $(3, 0, -2)$ .  $L_0$  (throat depth) is POSTULATED ACTION-geometry (like Stage 009's  $d$ , not a medium constant); zero new counted knobs.

**Verification.** Dual-engine, independent routes, zero file I/O — the source pair’s scratch-YAML/\_sympy\_exprs.wl export, the Mathematica-YAML re-read, and the `expression_digest/engine_agreement` plumbing are retired; the .wl derives the certificate, the  $L_s$  assembly, the speed extraction, the domain solve, and the dimensional leg natively (own D/Solve/Coefficient/Simplify), with an arity self-check and no silenced messages; the transfer matrix, if present, is only an operator-nature cross-check (no DtN/pole/Robin — Stage 012). SymPy 61 PASS / Mathematica 71 PASS (71 = 61 shared + 8 arity self-checks + 2 native trig-basis confirmations), exit 0, CWD-independent. Full tri-review on fresh agents: FIDELITY\_CLEAN (all values hand-re-derived; the three booleans confirmed genuinely PRODUCED; the 011/012 boundary clean) and ADVERSARIAL\_CLEAN (the de-rig proven able-to-fail — operator via  $M/N/B$ , speed via renormalization, domain via taper — and the BdG confirmed a real fourth-order term). Three teeth/label nits were remediated (the firewall tooth made genuine, a witness-path tooth added so the intrusion flag is independently load-bearing, the arity labels corrected) and re-verified REVERIFY\_CLEAN.

## B.7 Stage 012: DtN Pole Ladder & Robin Falsifier ( $Z_{00} = -(\omega/c_S) \tan$ ) (Check II-G1b)

**Part / anchor.** Part II – Gravity (brane↔bulk return; the frozen-throat resonator response; COMPLETES the pathA\_30 D/N fold opened by Stage 011)

**Claim status.** EXACT WITHIN CLOSURE

**Purpose.** On Stage 011’s cited frozen constant-coefficient Helmholtz operator  $L_s\psi = \psi'' + (\omega/c_S)^2\psi = 0$  on  $s \in [0, L_0]$ , pose and solve the Dirichlet/Neumann boundary-value problem: derive the outward-mouth Dirichlet-to-Neumann map  $Z_{00} = -(\omega/c_S) \tan(L_0\omega/c_S)$ , its half-shifted resonance ladder  $\omega_n = \pi c_S(n + \frac{1}{2})/L_0$ , its round-trip  $R_{rt} = 1$ , and the Robin counterfactual that proves the D/N determination is not hardcodable — landing the joint verdict at DN\_UNITTEST\_BC\_DEPENDENT because the boundary condition itself is imposed (`bc_derivation_emitted = false`, a banked calibration input).

**Status.** EXACT WITHIN CLOSURE — the DtN, ladder, round trip, and Robin counterfactual are exact algebra *within the imposed-BC closure*. This stage COMPLETES the joint pathA\_30 verdict DN\_UNITTEST\_BC\_DEPENDENT: Stage 011 carried the reduction-certificate component (REDUCTION\_CERTIFIED), and Stage 012 adds the D/N ladder plus the BC\_DEPENDENT landing. It is a near-pure *bridge-strip* (contrast Stage 011’s de-rig): the physics is already genuinely computed in the source and the .wl transfer-matrix route already carries it, so the reshape severs the scratch-YAML bridge and keeps the native content. The operator, domain, and speed are *consumed* from Stage 011 (dual-site citation-integrity); the stage opens at the general-solution `dsolve`. Its register contribution is the single *imposed* edge R28 (the D/N boundary is a banked calibration input), which discharges no knob.

**The D/N boundary-value problem and the DtN.** The `dsolve` of the cited  $L_s$  gives  $\psi = C_1 \sin(\omega s/c_S) + C_2 \cos(\omega s/c_S)$ . Imposing the Dirichlet mouth trace  $\psi(0) = \psi_M$  and the Neumann cap  $\psi'(L_0) = 0$ , the  $2 \times 2$  coefficient matrix (determinant  $-\omega \cos(L_0\omega/c_S)/c_S$ ) is solved by `LUsolve`, and the outward-mouth admittance is

$$Z_{00} = -\frac{\partial_s \psi|_0}{\psi|_0} = -\frac{\omega}{c_S} \tan\left(\frac{L_0\omega}{c_S}\right).$$

`dtn_matches_target` compares this *derived*  $Z_{00}$  (from the `LUsolve`) against the typed target  $-k \tan(kL_0)$  — a genuine derived-vs-typed comparison, not an  $X \equiv X$ : perturbing the Neumann RHS recomputes the derived DtN and the comparison fails.

**Ladder, static limit, round trip.** The DtN denominator’s  $\cos(L_0\omega/c_S)$  factor gives the pole equation  $\cos(L_0\omega/c_S) = 0$ , whose roots are the *half-shifted* ladder  $\omega_n = \pi c_S(n + \frac{1}{2})/L_0$ ; `halfshift` is computed as `pole_residual = 0`. The small- $\omega$  series  $-L_0\omega^2/c_S^2 - L_0^3\omega^4/(3c_S^4) + O(\omega^6)$  is the leading static compliance (distinct from the DC limit  $\lim_{\omega \rightarrow 0^+} Z_{00} = 0$ ; no separate static solve). With Dirichlet/Neumann reflections  $r_D = -1$ ,  $r_N = +1$ , the round-trip factor  $-\exp(2iL_0\omega/c_S)$  evaluates to  $R_{rt} = 1$  ( $\varphi_0 = 0 \bmod 2\pi$ ) on the ladder.

**The Robin falsifier.** Without a counterfactual the D/N determination is hardcodable. The Robin cap  $\partial_s\psi - \alpha\psi = 0$  interpolates Neumann ( $\alpha \rightarrow 0$ ) and Dirichlet ( $\alpha \rightarrow \infty$ ); its DtN reads off `robin_denominator_core = k cos(kL0) - α sin(kL0)`. The `counterfactual_guard` has exactly six members, each a *computed residual* (not a stamped truthiness): `robin_determinant_emitted` tied to the computed Robin determinant `robin_det ≡ -robin_denominator_core`; `recovers_DN_at_alpha0` ( $\alpha \rightarrow 0$  recovers  $-(\omega/c_S)\tan$ ); `recovers_DD_at_alpha_inf` ( $\alpha \rightarrow \infty$  recovers the D/D DtN  $k \cot(kL_0)$ ); `halfshift_destroyed_for_DD` (the D/D denominator  $\sin(kL_0)$  gives the *integer* ladder  $\pi c_S j/L_0$ , not the half-shifted one); `numeric_alpha_distinct` ( $\alpha = 2/L_0$  distinct from both limits, exact rationals); and `dtn_mismatch`. The artifact `dd_zero_mode_removable = 1/L0` is held out of the guard. Breaking `robin_denominator_core` flips the recomputed `recovers_DD_at_alpha_inf` and is caught ( $\rightarrow$ FAIL\_COUNTERFACTUAL).

**Imposed BC (honest scope) and the completed joint verdict.** The D/N boundary pair is IMPOSED (`bc_provenance = imposed`); the explicit mouth/cap  $V_{\text{wall}}$  gradient derivation is *not* emitted (`bc_derivation_emitted = false`), so the verdict lands at `DN_UNITTEST_BC_DEPENDENT` rather than `DN_UNITTEST_PASS` (edge R28; earning PASS is a deferred upgrade, not fabricated here). The 012-scoped verdict is a function of the computed rungs (`FAIL_DIMENSIONAL`  $\rightarrow$  `FAIL_POLE_LADDER`  $\rightarrow$  `FAIL_COUNTERFACTUAL`  $\rightarrow$  the `BC_DEPENDENT` landing), and the joint verdict is printed as `DN_UNITTEST_BC_DEPENDENT = (011: REDUCTION_CERTIFIED, cited) ∧ (012: DtN ladder + BC_DEPENDENT landing, computed here)`, not typed as 012-earned.

**Consumed + dimensional leg.** The frozen operator  $L_s$ , domain  $[0, L_0]$ , and speed  $c_S$  are CONSUMED from Stage 011 with dual-site citation-integrity: for  $L_s$ , site A the typed export form and site B the *null-space reconstruction* from the `dsolve`’d pair  $\{\sin, \cos\}$  (posit  $y'' + ay' + by$ , solve for  $(a, b) = (0, (\omega/c_S)^2)$ ) — a genuinely independent construction (a  $\sin/\cos \rightarrow \sinh/\cosh$  corruption recovers  $b = -(\omega/c_S)^2$  and fires); for  $c_S^2 = 5K\rho^4/m$  (R1 at  $\rho_*$ ), the literal vs the EOS route  $\partial_\rho(K\rho^5)/m$ . The 012 dimensional legs use the  $[K] = [P] - 5[\rho]$  chain:  $[\tan\_argument] = [kL_0] = 1$  (dimensionless),  $[Z_{00}\_prefactor] = [-k] = L^{-1}$ ; the corrupt- $[K]$  probe propagates  $[c_S^2] \rightarrow [c_S] \rightarrow [k]$ , flipping both legs  $\rightarrow$  `DN_UNITTEST_FAIL_DIMENSIONAL`.  $\alpha$  (Robin cap admittance,  $[\alpha] = L^{-1}$ ) is a control-construction symbol; zero new counted knobs.

**Verification.** Dual-engine, independent routes, zero file I/O — the source pair’s scratch-YAML/`_sympy_exprs.wl` export, the Mathematica-YAML re-read, and the `expression_digest/engine_agreement` plumbing are retired; the `.wl` KEEPS its transfer-matrix route as the independent engine (propagate the mouth state through `transferMatrix[L0]`, impose the Neumann cap, read  $Z_{00}$  — a different decomposition from the `.py` `LUsolve`, computed agreement), preserving its native

`robinAlpha0==dtnTransfer/robinAlphaInf==ddTransfer` self-checks, with an arity self-check and no load-bearing message silenced. SymPy 84 PASS / Mathematica 90 PASS (the +6 = the .w1 arity block net of two SymPy-only DtN-detail checks), exit 0, CWD-independent. Full tri-review on fresh agents: **FIDELITY\_CLEAN** (all values hand-re-derived; the six-member Robin guard each a genuine residual; `dtn_matches_target` genuine derived-vs-typed; the  $L_s$  null-space dual-site independent; the consume-from-011 boundary clean) and **ADVERSARIAL\_CLEAN** (16-mutant matrix — the guard un-stampable, the  $\sin/\cos \rightarrow \sinh/\cosh$  corruption fires in both engines, even a hardcoded DtN target is caught, all four verdict rungs reachable, arity clean). Eight able-to-fail teeth, mutations on copies.

## B.8 Stage 013: Breathing Harmonic Profiles & $M/K$ Operator Projection (Check II-G2a)

**Part / anchor.** Part II – Gravity (the frozen-throat  $\ell = 0$  breathing mode; the  $(a, L)$  collective-coordinate closure; the  $M/K$ -projection component of the joint **BREATHING\_CALIBRATED**, with truncation consistency = Stage 014 and the legacy-Hessian + Hellmann–Feynman force = Stage 015)

**Claim status.** EXACT WITHIN CLOSURE

**Purpose.** Reduce the  $\ell = 0$  linearized breathing of Gate-1’s frozen constant-coefficient wall operator  $\mathcal{L}_0 = \mu_\eta^{-1}(-\partial_w(T_w \partial_w \cdot) + K_\eta \cdot)$  on  $w \in [0, L_0]$  to a two-mode collective closure  $Q = (\delta_a, \delta_L)$ : solve the harmonic-lift profiles  $\alpha_a, \alpha_L$  (from  $\mathcal{L}_0 \alpha = 0$  under the collective BCs), project the wall inertia and stiffness onto them by *real  $\int dw$  operator projection* to build the mass/stiffness matrices  $M_{AB}, K_{AB}$ , and assemble the dynamical-EOM left-hand side  $M_{AB} \ddot{Q}^B + K_{AB} Q^B = F_A^{(\text{HF})}$ . The first Part-II *calibrated* stage: the structure is earned, the wall constants are calibration inputs.

**Status.** EXACT WITHIN CLOSURE — the harmonic-lift profiles, the  $M_{AB}/K_{AB}$  operator projection, and the EOM left-hand side are exact algebra *within the calibrated wall closure*. The verdict is the joint **BREATHING\_CALIBRATED**; this stage carries its  $M/K$ -projection +  $(a, L)$ -closure component. It is the **first Part-II calibrated stage**: the gravity-sector algebra was knob-free through Stages 008–012, and the calibration enters here with the breathing-mode wall response. *The structure is EARNED; the values are CALIBRATED*. The frozen throat packet is *consumed* from Gate-1 (Stages 011/012) with dual-site citation-integrity;  $c_S$  is *not* consumed (the matter-sector  $c_s/\text{BdG } k^4$  is deferred under  $k\xi \ll 1$ ).

**The harmonic-lift profiles.** The general solution of  $\mathcal{L}_0 \alpha = 0$  ( $\Leftrightarrow \alpha'' - \beta^2 \alpha = 0$ , with  $K_\eta = T_w \beta^2$ ) is  $\alpha = C_1 \sinh(\beta w) + C_2 \cosh(\beta w)$ . Under the collective BCs,

$$\alpha_a = \frac{\sinh(L_0 \beta - \beta w)}{\sinh(L_0 \beta)} \quad (\alpha_a(0) = 1, \alpha_a(L_0) = 0), \quad \alpha_L = \frac{r_{AL} \sinh(\beta w)}{\sinh(L_0 \beta)} \quad (\alpha_L(0) = 0, \alpha_L(L_0) = r_{AL}).$$

The harmonic-residual assertion  $\text{Lop}[\alpha] = (-T_w \alpha'' + K_\eta \alpha)/\mu_\eta \equiv 0$  (a raise on failure) proves the profiles genuinely annihilate the operator. Its able-to-fail tooth uses a *non-kernel* corruption (a wrong wavenumber  $\sinh(L_0 \beta - 2\beta w)$  or  $\alpha + 1$ ) — a mere rescale or argument sign-flip stays in  $\ker(\mathcal{L}_0)$  and leaves the residual 0, so it would be a non-tooth; the adversarial leg’s decisive ablation confirmed a kernel-preserving corruption is caught only by the *separate* BC assert.



**$M/K$  by real  $\int dw$  operator projection.** With  $\langle f, g \rangle_\mu = 4\pi \int_0^{L_0} \mu_\eta f g dw$ ,

$$M_{AB} = 4\pi \int_0^{L_0} \mu_\eta \alpha_A \alpha_B dw, \quad K_{AB} = 4\pi \int_0^{L_0} [T_w \alpha'_A \alpha'_B + K_\eta \alpha_A \alpha_B] dw,$$

computed by `sp.integrate` / native `Integrate`. Closed forms (both engines agreeing):  $K_{aa} = 4\pi T_w \beta / \tanh(L_0 \beta)$ ,  $K_{aL} = -4\pi T_w \beta r_{AL} / \sinh(L_0 \beta)$ ,  $K_{LL} = 4\pi T_w \beta r_{AL}^2 / \tanh(L_0 \beta)$ ,  $\det K = 16\pi^2 T_w^2 \beta^2 r_{AL}^2$ ; and the  $M_{AB}$  forms with  $[M_{AB}] = M$ ,  $[K_{AB}] = M T^{-2}$ . The genuineness spine (the anti-“typed  $M/K$ ” burden, since one could otherwise type the legacy static Hessian  $\{\chi^2 \kappa + \sigma_A, -\chi \kappa, \kappa + \sigma_L\}$ ) is twofold: (i) an *independent re-integration* of the full integrands compared against the hardcoded report closed-forms (not against the same projection call — that would be a vacuous  $X \equiv X$ ); dropping the  $T_w \alpha' \alpha'$  gradient makes it mismatch and fires. (ii) the `forbidden_fit_flags` (`K_from_static_hessian`, `M_or_K_typed_from_legacy_values`, `full_matrix_fit`) computed from the *symbol names* of  $M_{AB}/K_{AB}$  — robust to SymPy’s assumption-fragility (`Symbol('kappa') \neq Symbol('kappa', positive=True)`), so an object-set intersection would silently miss): the names must lie in  $\{\mu_\eta, T_w, \beta, L_0, r_{AL}, \pi\}$  and contain none of  $\{\kappa, \chi, \sigma_A, \sigma_L\}$ ; typing  $K_{AB}$  from the legacy Hessian introduces those names, flips the flag, and is caught in both engines (`K_AB_provenance=operator_projection_not_static_Hessian`).

**The dynamical-EOM left-hand side.** The collective EOM LHS is  $M_{AB} \ddot{Q}^B + K_{AB} Q^B = F_A^{(\text{HF})}$  with  $Q = (\delta_a, \delta_L)$ , emitted as the two expanded rows. This stage emits the LHS and EOM structure only; the right-hand side  $F_A^{(\text{HF})}$  (the Hellmann–Feynman force) is a symbolic placeholder filled by Stage 015 (as is the HF two-route enforcement, the `pathA_31 v1 rig locus`). A native  $M_{\text{det}} \rightarrow 0$  degenerate counterfactual ( $\alpha_a \rightarrow 0$ , recomputed via the same projection) is caught by a native `M_det \equiv 0` test, not the Stage-015 structure gate.

**Consumed packet and the anti-tautology guard.** The frozen wall packet  $\{L_0 = 37/20, T_w = 1, K_\eta = 1, \beta = 1\}$  is cited from Gate-1 with two independently-corruptible relations: site A (constitutive)  $\beta - \sqrt{K_\eta/T_w} \equiv 0$  and site B (branch)  $\beta L_0 - 37/20 \equiv 0$  (the branch-determinable  $L/a$ ), plus the frozen-export anchor. *The packet values are held as independent cited datums* (`K_eta_cited`, ...) — the guard does *not* re-expand  $K_\eta$  through the alias  $K_\eta = T_w \beta^2$ , which would collapse site A to  $\beta - \beta \equiv 0$  (a disguised  $X \equiv X$ ); a lone `K_eta_cited` corruption genuinely breaks site A.  $L_0$  is already registered (Stage 011, `ACTION-geometry`) — not re-counted.

**Calibration and register (the first Part-II counted knobs).** The wall constitutive packet contributes **three counted CALIB knobs**  $\{\mu_\eta$  (inertia,  $M L^{-1}$ ),  $T_w$  (tension,  $M L T^{-2}$ ),  $\beta$  (inverse-length scale,  $L^{-1}\}$ , with  $K_\eta = T_w \beta^2$  a *derived* manifestation (edge R29) — so three counted, not four.  $\beta$  is counted, *not* dressed as geometry: the source is explicit that geometry alone does not derive the wall constants, and  $\beta L_0 = 37/20$  is  $\beta_{(\text{calibrated})} \cdot L_0$  (= the  $L/a$  geometry), not an independent branch-pin of  $\beta$  (the conservative, non-shrinking read).  $r_{AL}$  (dimensionless collective BC ratio) is a control-construction symbol, tracked but not counted. Edge R30 is the named-pending nonlinear-throat reduction that would derive  $\{\mu_\eta, T_w, \beta\}$  from the medium (reduction debt).

**Verification.** Dual-engine, independent routes, zero file I/O — the source pair’s scratch-YAML/`_sympy_exprs.wl` export and the hybrid `.wl`’s `Get/sympy*-cross-check/Export` are retired; the `.wl` KEEPS its native `DSolveValue+Integrate` route but its asserts are *re-targeted* to its own native  $M/K$  (Mathematica-native `Csch/Coth` closed forms, algebraically equal to the SymPy `tanh/sinh` forms),

with a genuine arity self-check. The 3-way cut is clean: no Stage-014 truncation/generalized-eig/ $\beta_{L_0}$ -sweep and no Stage-015 legacy Hessian / Hellmann–Feynman force / static-dynamic limit is computed. SymPy 78 PASS / Mathematica 84 PASS (the +6 = the .w1 arity block), exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend; Grok’s compute-verification pass caught a kernel-preserving residual-tooth defect (folded) plus hardening nits. Full tri-review on fresh agents: FIDELITY\_CLEAN (every profile/ $M/K/\det$  independently re-derived via a different normalization path; `expand_hyperbolic_args` a math-preserving identity) and ADVERSARIAL\_CLEAN (a ten-ablation matrix, all firing: the free-symbol-name ancestry, the non-kernel residual tooth, the `K_eta_cited` anti-tautology, the native degenerate- $M_{\det}$  guard, the corrupt- $[T_w]$  probe). One non-blocking nit (the baseline re-integration was value-wise  $X \equiv X$ ) was remediated to compare against the hardcoded closed-forms, then REVERIFY\_CLEAN. Six able-to-fail teeth, mutations on copies.

## B.9 Stage 014: Truncation Consistency — Combined-Basis Generalized Eigenproblem, $\beta_{L_0}$ Sweep, and N-Convergence (Check II-G2b)

**Part / anchor.** Part II – Gravity (the frozen-throat  $\ell = 0$  breathing mode; the truncation-consistency component of the joint BREATHING\_CALIBRATED, with the  $M/K$ -projection +  $(a, L)$ -closure = Stage 013 and the legacy-Hessian + Hellmann–Feynman force = Stage 015)

**Claim status.** NUMERICALLY LOCATED

**Purpose.** Certify that Stage-013’s two-mode collective closure  $Q = (\delta_a, \delta_L)$  is a *faithful* truncation, not an artifact. Embed the two collective harmonic-lift profiles  $\alpha_a, \alpha_L$  in a larger Galerkin basis  $B = \{\alpha_a, \alpha_L, g_1 \dots g_N\}$  ( $N$  sine lane modes  $g_n = \sin((n - \frac{1}{2})\pi w/L_0)$ ), solve the combined-basis generalized eigenproblem  $Kv = \omega^2 Mv$ , and measure how much of each of the two lowest generalized modes lives in  $\text{span}\{\alpha_a, \alpha_L\}$  — the modal overlaps  $o_1, o_2$  — across a  $\beta_{L_0}$  sweep and as  $N \rightarrow 16$ . The first *numeric / float-bearing* stage of the rebuild.

**Status.** NUMERICALLY LOCATED — the generalized eigenproblem, the modal overlaps, the  $\beta_{L_0}$  validity window, and the N-convergence are numerically computed (`scipy.linalg.eigh+quad; Eigensystem+NIntegrate`). The verdict is the joint BREATHING\_CALIBRATED; this stage carries its *truncation-consistency* component (2/3). It **adds no new counted knobs**: the underlying wall constants  $\{\mu_\eta, T_w, K_\eta, \beta\}$  are Stage-013’s calibration inputs, and the numeric truncation controls are method/tolerance parameters. Stage-013’s collective profiles and the frozen packet are *consumed* with dual-site citation-integrity;  $c_S$  is *not* consumed (matter-sector deferred,  $k\xi \ll 1$ ).

**The combined-basis generalized eigenproblem.** Over  $B = \{\alpha_a, \alpha_L, g_1 \dots g_N\}$  with normalized  $x = w/L_0 \in [0, 1]$ , the mass/stiffness Gram is built by numeric quadrature,  $M_{ij} = \int_0^1 \phi_i \phi_j$ ,  $K_{ij} = \int_0^1 [\phi_i' \phi_j' + \beta^2 \phi_i \phi_j]$ , rank-deficient rows are dropped, and  $Kv = \omega^2 Mv$  is solved and sorted ascending. The two collective modes  $\alpha_a = \sinh(\beta(1-x))/\sinh \beta$ ,  $\alpha_L = \sinh(\beta x)/\sinh \beta$  are *cited* from Stage 013 (not re-derived); the numeric  $2 \times 2$  collective block equals Stage-013’s  $M_{AB}/K_{AB}$  only up to the  $4\pi$ ,  $L_0$ -Jacobian, and  $\mu_\eta$ -weight prefactors, which cancel in the generalized eigenproblem and the mass-normalized overlaps — so the seam is guarded via the consumed profiles, *not* a naive  $M_{aa}$  equality.

**The modal overlaps and the truncation floor.** Projecting each of the two lowest generalized eigenvectors onto  $\text{span}\{\alpha_a, \alpha_L\}$  via the mass-Gram gives  $o_k = \|P_\alpha v_k\|_M / \|v_k\|_M$ . The certificate is

$$\text{pass} \iff o_1 \geq \text{FLOOR} \wedge o_2 \geq \text{FLOOR} \wedge \min(\omega_1^2, \omega_2^2) > 0, \quad \text{FLOOR} = 0.9,$$

a genuine predicate on the *computed* overlaps. At the physical  $\beta_{L_0} = 37/20$ ,  $N = 16$ :  $o_1 = 0.99311$ ,  $o_2 = 0.98776$ ,  $\min(\omega^2) = 3.42252$ ,  $\text{gap} = 2.22787$ ,  $\text{pass} = \text{true}$ . All three conjuncts are separately able-to-fail (the  $o_2$  floor, the  $\min(\omega^2) > 0$  positivity leg, and the window edge).

**The computed validity window.** Sweeping  $\beta_{L_0} \in \{0.1, \dots, 3.0, 5, 10, 18, 30, 50\}$  — *spanning well beyond* the pass region (to  $o_1 \approx 0.27$  at  $\beta_{L_0} = 50$ ) — the certificate passes for  $\beta_{L_0} \in \{0.1 \dots 3.0\}$  and **fails** for  $\beta_{L_0} \geq 5$  ( $o_1 = 0.860 < \text{FLOOR}$  already at  $\beta_{L_0} = 5$ ). The window  $\{\text{min}:0.1, \text{max}:3.0\}$  is *computed* from the passing set; it has a real upper edge. The honest caveat: clean two-mode truncation holds only for order-unity wall stiffness  $K_\eta/T_w \lesssim 2.6$  (the emergent edge, since  $\beta^2 = K_\eta/T_w$  and  $(3/1.85)^2 \approx 2.63$ ) — an emergent number, not a typed criterion. The physical  $\beta_{L_0} = 37/20$  is the *cited* Gate-1 anchor, not the  $\beta$  that maximizes overlap. *The pathA\_31 v1 rejection was a gamed truncation threshold; the floor and window here are genuinely computed and able-to-fail.*

**N-convergence and the counterfactual overlaps.** At  $\beta_{L_0} = 37/20$ ,  $o_1$  is stable across  $N \in \{4, 8, 12, 16\}$  ( $0.99301 \rightarrow 0.99311$ ,  $\max|\Delta| \approx 10^{-4}$ ),  $\text{pass} = \text{true}$  at every  $N$  — so the certificate is not a small- $N$  artifact (the mass condition growth  $3.1\text{e}3 \rightarrow 1.9\text{e}5$  is a benign conditioning effect). Two counterfactual profiles pin the honest scope: the *degenerate*  $\alpha_a \equiv 0$  collapses the collective span (rank deficient,  $o_2 = 0.2227 < \text{FLOOR}$ ) and **fails** the floor; the *constant*  $\alpha_a \equiv 1$  — a *wrong* profile — gives  $o_1 = 1.0$ ,  $o_2 = 0.9738 \geq \text{FLOOR}$  and **passes** the overlap. *The overlap therefore certifies truncation-consistency, not profile-correctness*: Stage 014 records that the constant profile passes and does *not* claim it is rejected — its rejection is Stage-015’s Hellmann–Feynman mismatch (and Stage-013’s  $\mathcal{L}_0[\alpha] = 0$  residual).

**Consumed closure and register.** The collective profiles and the frozen packet  $\{L_0 = 37/20, \beta = 1, \beta_{L_0} = 37/20\}$  are cited from Stage 013 with two independently-corruptible relations: site A (branch anchor)  $\beta_{L_0} - 37/20 \equiv 0$ , reading an independent cited datum; and site B (profile-consumption) the consumed profiles satisfy *both* the harmonic residual  $-\alpha'' + \beta^2\alpha = 0$  *and* the collective BC values — the residual alone misses a kernel-preserving corruption (a rescale leaves the residual 0 but breaks the BC), so residual $\wedge$ BC are both able-to-fail. Stage 014 adds **zero new counted knobs**: the numeric controls  $\{\text{FLOOR} = 0.9, N_{\text{final}} = 16, N_{\text{conv}}, \beta_{L_0}\text{-sweep}\}$  are method/tolerance parameters (tracked, not counted); the wall packet is consumed from Stage 013 (no double-count). One structural edge, R31 (the truncation validity window  $K_\eta/T_w \lesssim 2.6$ ), discharges nothing.

**Verification.** Dual-engine, independent numeric routes, zero file I/O — the source pair’s numeric-WL scratch export and the hybrid .wl’s `Get/sympy*-float-diff/Export` are retired; the .wl **KEEPS** its native `NIntegrate+Eigensystem galerkinRow` route but its asserts are *re-targeted* to its own computed overlaps, floor, and window (it declares its own sweep grid and the cited profile closed forms natively, not via `DSolveValue`), with a genuine arity self-check. Agreement is transcript-level (both engines independently reach the same pass-pattern, window, and  $o_k$  to  $\sim 10^{-13}$ , tighter than the 0.9 margin). The 3-way cut is clean: no Stage-013  $\int dw$  projection and no Stage-015 Hellmann–Feynman force is computed. SymPy/SciPy 93 PASS / Mathematica 100 PASS (the +7 = the .wl arity/leakage block), exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend (three BLOCKING folds: the site-B residual $\wedge$ BC, the  $\min(\omega^2) > 0$  tooth, and the N-convergence

non-converging ablation; Grok’s compute-verification independently re-derived every anchor). Full tri-review on fresh agents: FIDELITY\_CLEAN (an independent from-scratch Galerkin re-derived every value) and ADVERSARIAL\_ISSUES — the anti-gaming floor/window proven clean (23/23 genuine ablations firing), but *two* able-to-fail teeth (the constant-passes-overlap honesty leg and the degenerate-bypass leg) were vacuous  $x:=\text{const}$ ; `expect_fail(x)` stamps in both engines. Remediated to a *shared predicate* read by both the baseline and the tooth’s mutated-copy assertion; a fresh-agent re-verify confirmed REVERIFY\_CLEAN via the deciding coupling meta-test (corrupting the shared predicate makes the tooth stop firing, so the audit correctly fails). Nine able-to-fail teeth, mutations on copies.

## B.10 Stage 015: Legacy-Hessian Structure Recovery & the Hellmann–Feynman Force (Check II-G2c)

**Part / anchor.** Part II – Gravity (the frozen-throat  $\ell = 0$  breathing mode; the legacy-structure + Hellmann–Feynman-force component of the joint BREATHING\_CALIBRATED, which it *completes* — with the M/K-projection +  $(a, L)$ -EOM-LHS = Stage 013 and the truncation consistency = Stage 014)

**Claim status.** EXACT WITHIN CLOSURE

**Purpose.** Complete the calibrated breathing sector. **(Q1)** Show Stage-013’s operator-projected closure  $M_{AB}, K_{AB}$  has the same structural signature as the legacy static energy Hessian  $H_{\text{legacy}} = \partial^2 E_{\text{geom}} / \partial Q^2$  (the  $(a, L)$  closure *recovered*, not re-postulated). **(Q2)** Supply the EOM driving force  $F_A^{(\text{HF})}$  — the right-hand side Stage 013 deferred — by *two genuinely different* constructions (a distributed source projection and a Hellmann–Feynman parametric derivative) that agree only after simplification. Then take the static-dynamic limit and land the completed joint verdict.

**Status.** EXACT WITHIN CLOSURE — the structure recovery and the Hellmann–Feynman force are exact closed-form algebra (`sp.integrate/FullSimplify`; `expect_zero` residuals, exact rationals  $L_0 = 37/20$ ; the symbolic / float-free inverse of the numeric Stage 014). The verdict is the joint BREATHING\_CALIBRATED; this stage carries its *legacy-structure + HF-force* component (3/3) and **completes** it, filling the EOM RHS  $F_A^{(\text{HF})}$ . It adds **one** new counted CALIB knob — the breathing driving-force scale  $V_{p0}/\ell_c$  (with  $V_{p0}$  and the now-live  $\ell_c$  its manifestations;  $\rho_*$  consumed) — making it the second calibration-adding Part-II stage after Stage 013. Stage-013’s profiles and  $M_{AB}/K_{AB}$  and Stage-014’s truncation certificate are *consumed* with dual-site citation-integrity; the legacy constants  $\{\kappa, \chi, \sigma_a, \sigma_L\}$  are the legacy-Hessian pattern basis (not counted afresh);  $c_S$  is *not* consumed (matter-sector deferred,  $k\xi \ll 1$ ).

**The legacy energy and its Hessian.** With  $E_{\text{geom}} = \frac{1}{2}\kappa(\delta_L - \chi\delta_a)^2 + \frac{1}{2}\sigma_a\delta_a^2 + \frac{1}{2}\sigma_L\delta_L^2$ ,

$$H_{\text{legacy}} = \frac{\partial^2 E_{\text{geom}}}{\partial Q^2} = \begin{pmatrix} \chi^2\kappa + \sigma_a & -\chi\kappa \\ -\chi\kappa & \kappa + \sigma_L \end{pmatrix},$$

symmetric, negative off-diagonal ( $-\chi\kappa < 0$  for  $\chi, \kappa > 0$ ),  $\det = \kappa\sigma_a + \kappa\chi^2\sigma_L + \sigma_a\sigma_L > 0$ . The legacy constants enter *here*; a firewall echo confirms the cited  $M_{AB}/K_{AB}$  free symbols exclude them.

**The structure recovery.** The gate reads the cited  $M_{AB}/K_{AB}$  and matches the structural signature of  $H_{\text{legacy}}$ :  $M$  positive-definite by Sylvester —  $M_{aa} > 0$  and  $\det M > 0$  for  $B = \beta L_0 > 0$  — discharged by *exact-identity certificates* (not `is_positive`, which returns `None` on the transcendentals, and *not* the  $M_{aa} - m_{aa}^+ = 0$  form-equality, which is  $X \equiv X$  once  $M$  is cited):  $f_1 = \sinh B \cosh B - B$  has  $f_1(0) = 0$ ,  $f_1' = \cosh 2B - 1 = 2 \sinh^2 B \geq 0$ ;  $f_2 = \sinh^2 B - B^2 = (\sinh B - B)(\sinh B + B) > 0$  via  $g = \sinh B - B$ ,  $g' = \cosh B - 1 = 2 \sinh^2(B/2)$ . Then  $K$  symmetric,  $K_{aL} < 0$ ,  $\text{rank } K = \text{rank } H_{\text{legacy}} = 2$ , and matching zero-pattern give  $K_{\text{structure ok}}$ ; full matrix fit = false (the recovery is structural,  $M_{aa} \neq H_{aa}$ ). The probes  $M_{aa} \rightarrow -M_{aa}$  and  $K_{aL} \rightarrow -K_{aL}$  flip the gate. *The operator-projected closure has the same structural form as the legacy energy Hessian.*

**The Hellmann–Feynman force (both routes).** With the confinement force density  $\text{src} = \rho_* V_{p0}/\ell_c$  and the action measure  $4\pi \int dw$ :

$$\underbrace{F_A^{\text{dist}} = 4\pi \int_0^{L_0} \text{src } \alpha_A dw}_{\text{distributed projection}} = \underbrace{F_A^{\text{leg}} = -4\pi \int_0^{L_0} \rho_* \left. \frac{\partial V_{\text{conf}}}{\partial q_A} \right|_0 dw}_{\text{Hellmann–Feynman parametric}}$$

$V_{\text{conf}} = (V_{p0}/\ell_c)(r - R_0(w) - q_a \alpha_a - q_L \alpha_L)$ , Route 2 a genuine  $\partial/\partial q$  (the double-negative). They agree (hf force reduces = true,  $F_a^{\text{dist}} = 4\pi V_{p0} \rho_* (\cosh L_0 \beta - 1)/(\beta \ell_c \sinh L_0 \beta)$ ), yet the *raw* unsimplified trees are computed-distinct: unsimplified routes identical = false ( $+4\pi f(+\dots)$  vs  $-4\pi f(-\dots)$ ). *This is the pathA\_31 v1 rejection's other locus — the HF  $x-x$  tautology — emitted here as two genuinely different constructions; the verdict rung places the anti- $x-x$  in the OR: typing Route 2 as a copy of Route 1 trips BREATHING\_FAIL\_HF\_FORCE.* This fills the deferred RHS  $F_A^{(\text{HF})}$ .

**Static-dynamic limit and the profile guard.**  $\ddot{Q} \rightarrow 0$  in  $M_{AB}\ddot{Q} + K_{AB}Q = F_A$  gives  $K_{AB}Q = F_A$ , the static  $\partial E_{\text{geom}}/\partial Q = 0$  limit (the exact  $\ddot{Q} \rightarrow 0$  EOM residual identity, not a separate solve). The HF guard rejects both wrong profiles ( $\alpha_a \equiv 0$ :  $F_a^{\text{dist}} = 0$ ;  $\alpha_a \equiv 1$ :  $F_a^{\text{dist}} = 4\pi L_0 V_{p0} \rho_*/\ell_c$ ; both  $\neq F_a^{\text{leg}}$ , so hf mismatch = true). *The 014↔015 boundary:* the constant profile *passes* Stage-014's modal overlap but *fails* Stage-015's HF — the Hellmann–Feynman mismatch is the profile guard the overlap could not supply (with Stage-013's  $\mathcal{L}_0[\alpha] = 0$  residual).

**Consumed closure and register.** Stage-013's profiles,  $M_{AB}/K_{AB}$ , and the frozen packet  $\{L_0 = 37/20, \beta = 1, \beta L_0 = 37/20\}$  are cited with dual-site integrity: site A the branch anchor  $\beta L_0 - 37/20 \equiv 0$ ; site B the profiles satisfy *both* the harmonic residual  $-\alpha'' + \beta^2 \alpha = 0$  *and* the BC values (residual alone misses a kernel-preserving corruption); and the  $M/K$  citation covers every entry — two det-identities for the diagonals, and, because det carries  $M_{aL}^2$  (blind to an off-diagonal sign flip), explicit sign checks  $M_{aL} > 0$  (via  $h = B - \tanh B$ ) and  $K_{aL} < 0$ . Register: **one** new counted CALIB knob, the driving-force scale  $V_{p0}/\ell_c$ ; the legacy  $\{\kappa, \chi, \sigma_a, \sigma_L\}$  are the pattern basis (not counted afresh). Two structural edges: R32 (the legacy-Hessian structure recovery, discharges nothing) and R33 (the confinement-drive reduction debt, a sibling of R30's wall-response debt, PENDING).

**Verification.** Dual-engine, independent symbolic routes, zero file I/O — the source pair's `sympy*` export and the hybrid `.wl's Get/checks/Export/digest` are retired; the `.wl` KEEPS its native HF `Integrate`, Hessian D, and structure `Det/MatrixRank`, re-targeted to its own values, citing the profiles and  $M/K$  as native literals (not `DSolveValue`, not `re-Integrate`) and building its own  $H_{\text{legacy}}$ , HF routes, and raw-tree anti- $x-x$  comparison, with an arity self-check. Agreement is transcript-level (the `.wl`  $\tanh(\beta L_0/2)$  form equals the `.py`  $(\cosh L_0 \beta - 1)/\sinh L_0 \beta$  via the half-angle identity). The 3-way cut is clean: no Stage-013  $\int dw$  projection, no Stage-014 generalized eig. SymPy 95 PASS / Mathematica

102 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend (two BLOCKING folds: the det-blind off-diagonal sign hole  $M_{aL} \rightarrow -M_{aL}$ , added explicit sign checks; and the float-free positivity-certificate requirement —  $M_{aa} > 0$  is not dischargeable by `is_positive`, so exact-identity certificates replace the vacuous form-equality). Full tri-review on fresh agents: **FIDELITY\_CLEAN** (an independent read hand-verified all seven certificate identities and every value) and **ADVERSARIAL\_ISSUES** — the three central burdens proven genuine by live copy-mutations, but *one* able-to-fail tooth (the HF profile-guard leg) was a vacuous  $x-x$  self-compare in both engines. Remediated to read the real wrong-profile forces (with three lower-severity literal/scaffolding flags made-genuine or honestly de-counted); a fresh-agent re-verify confirmed **REVERIFY\_CLEAN** via the coupling meta-test (each remediated construct fires under a mutation of its named object). Eight able-to-fail teeth, mutations on copies.

## B.11 Stage 016: The $\ell = 2$ SO(3) Covariance Theorem (Check II-G3a)

**Part / anchor.** Part II – Gravity (the frozen-throat  $\ell = 2$  sector; the EARNED-FIRST leg of a two-way split of pathA\_32 — the SO(3) covariance-theorem component of the joint **ISOTROPY\_CALIBRATED**, with the grouped-P2 lane isotropy + the calibration partition = Stage 017)

**Claim status.** EXACT WITHIN CLOSURE

**Purpose.** Earn the  $\ell = 2$  angular structure the grouped-lane isotropy rides. Show the five real  $\ell = 2$  harmonics form one orthonormal SO(3)-irrep (Gram =  $I_5$ ) with a single *computed*  $-\Delta_{S^2}$  eigenvalue  $\lambda_m = \ell(\ell + 1) = 6$  for every  $m$  (Laplace–Beltrami + Rayleigh quotient + eigenfunction residual), and that the  $K_2$  angular stiffness  $K_2 = \tilde{K} + \lambda_m \tilde{T}_\Omega$  uses that computed eigenvalue. Land the joint verdict as a partial component.

**Status.** EXACT WITHIN CLOSURE — the covariance theorem is exact symbolic algebra (genuine  $S^2$ -integral Gram, the real  $-\Delta_{S^2}$  operator, Rayleigh quotients, eigenfunction residuals, exact  $\{L, M, T\}$  dimension vectors; `expect_zero/expect_bool` residuals, no `scipy/numpy/` floats/tolerances). The verdict is the joint **ISOTROPY\_CALIBRATED**; this stage carries its *SO(3) covariance-theorem* component (1/2, the EARNED slice) and lands the joint as **partial** (016 earned, 017 pending) — not complete. It adds **zero** new counted knobs (a structural covariance edge, like Stages 011/012/014); the new symbols first appearing here —  $T_\Omega/\tilde{T}_\Omega$  (the angular-stiffness density) and  $\beta_2(w)$  (the frozen  $\ell = 2$  radial profile) — have their counting *deferred* to Stage-017’s calibration partition.  $c_S$  is *not* consumed (the angular structure is speed-free).

**The real basis and orthonormality.** The five real  $\ell = 2$  harmonics  $Y_{20} = \sqrt{5/16\pi} (3 \cos^2 \theta - 1)$ ,  $Y_{21c} = -\sqrt{15/4\pi} \sin \theta \cos \theta \cos \phi$ ,  $Y_{21s} (\cos \phi \rightarrow \sin \phi)$ ,  $Y_{22c} = \sqrt{15/16\pi} \sin^2 \theta \cos 2\phi$ ,  $Y_{22s} (\cos 2\phi \rightarrow \sin 2\phi)$  are orthonormal on the sphere:

$$\text{Gram}_{ij} = \int_{S^2} Y_i Y_j d\Omega = \int_0^\pi \int_0^{2\pi} Y_i Y_j \sin \theta d\phi d\theta = \delta_{ij}, \quad \text{Gram} = I_5.$$

This is a computed integral — a wrongly-normalized harmonic breaks gram is identity and drives *covariant* to **FAIL\_NOT\_COVARIANT**.

**The computed eigenvalue (the crux).** With the real Laplace–Beltrami operator  $-\Delta_{S^2} f = -[(1/\sin \theta)\partial_\theta(\sin \theta \partial_\theta f) + (1/\sin^2 \theta)\partial_\phi^2 f]$ , two independent computations certify  $\lambda_m$  is *computed*, not typed: the Rayleigh quotient  $\lambda_m = \int Y_m(-\Delta_{S^2})Y_m d\Omega / \int Y_m^2 d\Omega = 6$  for every  $m$ , *and* the eigenfunction residual  $(-\Delta_{S^2})Y_m - \lambda_m Y_m = 0$  (proving each  $Y_m$  is a genuine eigenfunction). So  $\lambda_m = \ell(\ell + 1) = 6$  is the same across all five  $m$ : the  $\ell = 2$  sector is one 5-dimensional SO(3)-irrep with a single eigenvalue — the angular degeneracy that *is* the covariance.

**The  $K_2$  angular stiffness.** The angular stiffness  $K_2 = \tilde{K} + \lambda_m \tilde{T}_\Omega$  (with  $M_2 = \tilde{M}$ ) is assembled with the *live* computed eigenvalue (`build_K2(lambdas)`, not a typed 6). Its computed-ness is able-to-fail through a residual on the assembled coefficient  $((-\Delta_{S^2})Y - (\text{coeff in } K_2)Y = 0$ , which a wrong typed coefficient breaks) *and* the *bare* forced eigenvalue probe (coefficient  $2 \neq 6 \Rightarrow \text{FAIL\_NOT\_COVARIANT}$ ). The source’s  $k_{\text{coeff equal}}$  self-compare  $\lambda - \lambda \equiv 0$  is *vacuous and is de-counted*; its role is carried by these two genuine checks.

**Angular dimensional consistency.** On pathA\_32’s own convention — volume densities on the wall measure  $dV = a^2 dw d\Omega$  (dimension  $L^3$ ),  $\beta_2$  dimensionless,  $\beta_2' = L^{-1}$ ,  $[\mu_\eta] = ML^{-3}$ ,  $[T_w] = ML^{-1}T^{-2}$ ,  $[K_\eta] = [T_\Omega] = ML^{-3}T^{-2}$  — one finds  $[M_2] = M$ , each  $K_2$  term  $MT^{-2}$ ,  $[K_2/M_2] = T^{-2}$ . The dimensions are *sourced* from the density integrals (not back-solved). Corrupting the sourced  $[T_\Omega]$  (and  $\tilde{T}_\Omega$ ) by one power of  $L$  makes the  $K_2$  sum inhomogeneous  $\Rightarrow \text{FAIL\_DIMENSIONAL}$ ; the same holds for a one-power corruption of  $[\mu_\eta]$ ,  $[T_w]$ , or  $[K_\eta]$ .

**The able-to-fail battery.** The scoped verdict runs 016’s gates only (dimensional, covariant, tautology, able-to-fail; 017’s stability/denominator/lane/dynamic gates are not computed here). The aggregate battery is 016’s three probes — wrong eigenvalue (coeff  $2 \neq 6 \Rightarrow \text{FAIL\_NOT\_COVARIANT}$ ), tautology hash collision (identical harmonic inputs  $\Rightarrow$  non-distinct SHA-256 hashes  $\Rightarrow \text{FAIL\_TAUTOLOGICAL}$ ), and dimensional corrupt  $T_\Omega$  ( $\Rightarrow \text{FAIL\_DIMENSIONAL}$ ) — each a computed conjunction of (its mutation fires its verdict)  $\wedge$  (its self-ablation suppresses); *neutering any one flips the aggregate false*.

**Honest scope.** Angular earned / radial calibrated: 016 derives the angular structure, but the radial profile  $\beta_2(w)$  and the radial/support scalars remain frozen calibration inputs (not derived from the Gate-1  $R_0$  support equation) — which is *why* the joint is ISOTROPY\_CALIBRATED rather than ISOTROPY\_PASS; the calibration partition is Stage-017’s. The 54/5 quadrupole normalization, outgoing odd- $N$  coefficients, and the solved nonlinear branch ( $G = \text{GENUINE\_BLOCKED}$ ) are deferred Gate 4/5/6 work.

**Consumed closure and register.** *Provenance, not a cross-stage relation:* pathA\_32’s volume-density / dimensionless- $\beta_2$  convention differs from Stage-013’s line-density /  $\beta = L^{-1}$  convention (related by an  $\int a^2 d\Omega \approx L^2$  bridge, not equal), so the Stage-013 relation  $K_\eta = T_w \beta^2$  does *not* transfer. Hence  $\mu_\eta, T_w$  are cited as the same physical wall constants counted at Stage 013 (provenance), the frozen radial reference and the Gate-1 D/N boundary (Stage 012 R28) are cited, and the genuine able-to-fail integrity is pathA\_32’s *own*  $[M_2] = M/[K_2] = MT^{-2}$  dimensional consistency. Register: **zero** new counted knobs; the new  $T_\Omega/\beta_2$  counting is deferred to Stage 017; structural edge **R34** (the  $\ell = 2$  SO(3) covariance provenance — five real harmonics = one SO(3)-irrep with  $\lambda_m = 6$ ; discharges nothing).

**Verification.** Dual-engine, independent symbolic routes, zero file I/O — the source pair’s scratch-YAML engine-agreement handoff is severed; the .w1 KEEPS its native **Integrate/ D/Hash** route (its own `dimOf/scopedVerdict`), not a mirror, agreeing at the transcript level (Gram =  $I_5$ ,  $\lambda_m = 6$ , the three probe verdicts, the 016 aggregate), with an arity self-check. SymPy 82 PASS / Mathematica 91 PASS,

exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend; the Grok compute-verify pass caught the volume-vs-line dimensional-convention mismatch (the cross-stage dual-site was non-transferable → reframed to provenance + self-contained integrity), and strengthened the  $k_{\text{coeff equal}}$  de-count to a residual on the assembled  $K_2$  coefficient. Full tri-review on fresh agents: **FIDELITY\_CLEAN** (an independent read hand-re-derived the harmonics, the Laplace–Beltrami operator, all Rayleigh  $\lambda = 6$  with zero residuals, and the full  $L/M/T$  dim walk) and **ADVERSARIAL\_CLEAN** (per-tooth ablation: all 82 SymPy checks + 4 native .wl ablations fired at their own assert) — with one low-severity stamped literal (participates in verdict) remediated to a computed verdict-propagation check; a fresh-agent **REVERIFY\_CLEAN** confirmed it fires under its own ablation. Symbolic per-tooth ablation, mutations on copies.

## B.12 Stage 017: Grouped-P2 Lane Isotropy (Check II-G3b)

**Part / anchor.** Part II – Gravity (the frozen-throat  $\ell = 2$  sector; the COMPLETING leg of the two-way split of pathA\_32 — the grouped-lane isotropy + calibration partition that consumes Stage 016’s covariance theorem, lands the joint **ISOTROPY\_CALIBRATED**, and exports the  $\ell = 2$  port kernel)

**Claim status.** EXACT WITHIN CLOSURE

**Purpose.** Show that the grouped  $\{20, 21, 22\}$   $\ell = 2$  lanes, assembled from Stage 016’s consumed covariance ( $\lambda_m = 6$ ,  $K_2 = \tilde{K} + \lambda_m \tilde{T}_\Omega$ ), respond isotropically — the  $P_2$ -projection raw-D defects vanish (primary) and the normalized-u response defects vanish (cross-check) — and that the response is genuinely able-to-fail (the six-probe battery). Land the joint verdict as complete and export the port kernel.

**Status.** EXACT WITHIN CLOSURE — mixed exact-symbolic + numeric-sample: the lane algebra, D-lanes, raw-D/normalized-u defects, and the c/s degeneracy are exact symbolic (`compact(expr)==0`, `expect_zero/expect_bool`); the response-probe “moves under  $\varepsilon$ ” tests and the stability/denominator windows are numeric over a calibration sample/window (tolerance-guarded, as the source). The verdict is the joint **ISOTROPY\_CALIBRATED**; this stage carries its *grouped-lane isotropy* component (2/2) and lands the joint as **complete** (016 earned-cited  $\wedge$  017 earned-here). It is the *third* calibration-adding Part-II stage — it adds two counted **CALIB** knobs  $\{T_\Omega, \beta_2\}$ .

**The grouped-lane assembly.** Each channel is assembled from the consumed eigenvalue  $\lambda_m = 6$  and the  $K_2$  form:

$$M_2 = c_{\text{self}} \tilde{M}, \quad K_2 = c_{\text{self}} (\tilde{K} + \lambda_m \tilde{T}_\Omega), \quad c_{\text{self}} = \int Y_m^2 d\Omega = 1 \text{ (the Gram diagonal, cited from 016),}$$

with D-lanes  $D_0 = K_2 - (\tilde{B}_0 + \tilde{Z}_0)$ ,  $D_2 = -(\tilde{M} + \tilde{B}_2 + \tilde{Z}_2)$ ,  $D_4 = -(\tilde{B}_4 + \tilde{Z}_4)$ . The three grouped lanes are 20, 21 = avg(21c, 21s), 22 = avg(22c, 22s); this grouped  $\{M_2, K_2, D_{0,2,4}, \tilde{B}/\tilde{Z}\}$  is the exported  $\ell = 2$  port kernel.  $\lambda_m = 6$  and the  $K_2$  form are cited from Stage 016, not re-derived.

**Raw-D defects = 0 — the primary isotropy test.** With the  $P_2$ -projection defect\_pair( $t$ ) =  $((2t_{20} - t_{21} - t_{22})/10, (t_{21} - t_{22})/2)$ , the grouped D-lane defects vanish at every order,  $\{a_{D0}, b_{D0}, a_{D2}, b_{D2}, a_{D4}, b_{D4}\} = 0$  (exact), with the within-group degeneracy  $D_{21c} = D_{21s}$ ,  $D_{22c} = D_{22s}$ . This is the isotropy statement: the grouped  $\ell = 2$  response is angularly degenerate because the lanes ride 016’s single eigenvalue  $\lambda_m = 6$ .



**Normalized-u defects = 0 — the cross-check.** The DtN-observable normalized responses  $u_2 = -D_2/D_0$ ,  $u_4 = (D_2^2 - D_0 D_4)/D_0^2$  also have vanishing  $P_2$ -projection defects. This is a *cross-check*, not the primary test: a pure-prefactor anisotropy *moves* the raw-D defects but leaves the normalized-u defects zero (the pure prefactor probe), so normalized-u alone would miss a real anisotropy — raw-D is decisive.

**The consumed covariance — a genuine cross-stage dual-site.** Because Stages 016 and 017 share the same pathA\_32 convention (volume densities, dimensionless  $\beta_2$ ) — unlike 016’s provenance-only cite of Stage 013 — the cited  $\lambda_m = 6$  and  $K_2$  form a *checkable* relation. Site A: the  $\lambda_m$  fed into the lane assembly equals 016’s exported eigenvalue. Site B: the *assembled* lane  $K_2$ ’s  $\tilde{T}_\Omega$  coefficient equals the cited  $\lambda_m$  (reads the assembled lane, guarding the assembly formula). The  $\lambda$ -dimensionless trap: a corruption  $\lambda : 6 \rightarrow 4$  is dimensionally silent, keeps the windows positive, and *keeps raw-D isotropy incidentally unbroken* (all-lanes  $\lambda = 4$  gives raw-D= 0) — so an incidental gate cannot catch it. A one-site corruption fires an explicit integrity guard; the coordinated-both-sites corruption is caught by a single-harmonic  $(-\Delta_{S^2})Y_{20} = \lambda Y_{20}$  echo. Stage 017 does not re-derive the covariance theorem.

**The able-to-fail battery.** 017’s aggregate is its six response probes, each a computed conjunction; *neutering any one flips* able to fail ok *false*: pure prefactor/sector selective/ m dependent ( $\rightarrow$ FAIL\_ANISOTROPIC\_BRANCH, load-bearing on the computed move-flags — their forced verdict via `case_verdict(lane_collapse_ok=False)` is vacuous by design), degenerate  $\beta_2$  ( $\rightarrow$ FAIL\_STABILITY), singular denominator ( $\rightarrow$ FAIL\_SINGULAR\_RESPONSE), static drop inertia ( $\rightarrow$ FAIL\_STATIC\_RESPONSE). The three 016-owned probes are cited via 016’s three-probe aggregate; the joint able to fail is 017’s six  $\wedge$  016’s cited three.

**Honest scope and register.** Calibrated not PASS: 017 earns the grouped-lane isotropy, but the wall profile  $\beta_2(w)$ ,  $R_0(w)$ , the radial scalars  $\tilde{M}/\tilde{K}/\tilde{T}_\Omega$ , the support scalars  $\tilde{B}/\tilde{Z}$ , and the calibration window are frozen calibration inputs — *not* derived from the Gate-1  $R_0$  support equation (edge **R36**, the ISOTROPY\_PASS target). Register: two new counted CALIB knobs  $\{T_\Omega, \beta_2\}$  (the  $\ell = 2$  angular-stiffness density — a genuinely new physical DOF absent from the  $\ell = 0$  breathing packet — and the frozen  $\ell = 2$  radial profile). The radial scalars  $\tilde{M}/\tilde{K}/\tilde{T}_\Omega$  are *derived* manifestations  $= \int \rho \beta_2^2 dV$  (edge **R35**, the  $\ell = 2$  analogue of R29). The port-kernel support/Maxwell scalars  $\{\tilde{B}, \tilde{Z}\}$  are frozen inputs first load-bearing here but tracked/downstream-pinned (the isotropy verdict is value-independent of them; the  $\tilde{Z}$  Maxwell couplings pin at Stages 018–024).  $\mu_\eta, T_w$  are cited provenance (Stage 013;  $K_\eta = T_w \beta^2$  R29 non-transferable across the volume-vs-line convention);  $c_S$  is not consumed. Edge **R34** (016’s covariance provenance) is backfilled. Part-II counted CALIB set  $= \{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$ . Deferred (Gate 4/5/6): the 54/5 normalization, outgoing odd- $N$  coefficients, and the solved nonlinear branch ( $G$ =GENUINE\_BLOCKED).

**Verification.** Dual-engine, independent symbolic + native routes, zero file I/O — the source pair’s scratch-YAML handoff is severed; the .wl keeps its native Integrate/FullSimplify/ D/N route (its own `verdictFromGates/caseVerdict/dimOf`), not a mirror, agreeing at the transcript level (the grouped lanes, raw-D= 0, normalized-u= 0, the six probe verdicts, the joint ISOTROPY\_CALIBRATED), with an arity + unevaluated-leakage self-check. SymPy 118 PASS / Mathematica 127 PASS, exit 0, CWD-independent. The directive design-review used the Codex $\rightarrow$ Grok $\rightarrow$ Codex bookend; the Grok compute-verify pass caught a blocking gap (the harmonics/intS2 infrastructure 017 needs for its anisotropy coefficients had been mislabeled for exclusion) plus two nits (the export coefficient pinned to  $\tilde{K} + 6\tilde{T}_\Omega$ ; the three anisotropy probes’ forced verdict flagged vacuous so the load rides the move-flags), and independently confirmed that all-lanes  $\lambda = 4$  keeps raw-D= 0 — proving the explicit dual-site necessary. Full tri-review on fresh agents:

FIDELITY\_CLEAN (independent re-derivation of the anisotropy coefficients  $\{2/7, 1/7, -2/7\}$ , the  $Y_{20}$  echo, the pure-prefactor discriminator, the six probe verdicts, and the joint landing) and ADVERSARIAL\_ISSUES (42 per-tooth ablations, 41 firing at their own assert) with one genuine finding — the .w1 Site-B ( $K_2$ -form) residual reconstructed  $K_2$  from `lambdaByChannel` instead of reading the assembled lane, so an assembly-formula corruption fired only downstream. Remediated: the .w1 Site-B now reads the assembled lane (mirroring SymPy), with matching assembly-formula ablation teeth in both engines. A fresh-agent REVERIFY\_CLEAN confirmed the coupling meta-test (the fixed tooth fires at its own named assert on an assembly-formula corruption in both engines; neutering the fix makes it vacuous), with no regression. Symbolic/numeric per-tooth ablation, mutations on copies.

### B.13 Stage 018: The Outgoing $\ell = 2$ DtN Hankel Fingerprint and $\chi_Q$ Sign (Check II-G4a)

**Part / anchor.** Part II – Gravity (the frozen-throat  $\ell = 2$  radiative sector; the EARNED-FIRST leg of a four-way split of pathA\_33 — the outgoing DtN Hankel-fingerprint +  $\chi_Q$ -sign component of the joint QUAD\_CALIBRATED, with the prefactor algebra = Stage 019, the  $54/5 = 2 \cdot 27/5$  provenance partition + the CALIBRATED label = Stage 020, and the  $\hat{\mu}_0$ -free dimensional closure = Stage 021)

**Claim status.** EXACT WITHIN CLOSURE

**Purpose.** Earn the exterior-wave radiative signature the quadrupole normalization rides. Show the outgoing  $\ell = 2$  DtN response  $\hat{Y}_2^{\text{out}}(z) = -3/\Lambda_2^{\text{out}}(z)$ ,  $\Lambda_2^{\text{out}} = z h_2^{(1)'} / h_2^{(1)}$ ,  $z = a\omega/c_s$ , yields the DERIVED rational fingerprint  $u_2 = a^2/9c_s^2$ ,  $u_4 = 4a^4/81c_s^4$ ,  $v_5 = a^5/27c_s^5$  (the 27 computed from  $v_5^z = 1/27$ ), and the sign  $\chi_Q = +1$  outgoing /  $-1$  incoming computed from  $j_2 \pm i y_2$ . Land the joint verdict as a partial component.

**Status.** EXACT WITHIN CLOSURE — the fingerprint is exact symbolic algebra (the outgoing  $\ell = 2$  spherical-Hankel logarithmic derivative and its small-frequency series are exact expansions; `expect_zero/expect_bool/expect_fail` residuals, no `scipy/numpy/floats/tolerances`). The verdict is the joint QUAD\_CALIBRATED; this stage carries its *outgoing-fingerprint* +  $\chi_Q$ -sign component (1/4, the EARNED-FIRST slice) and lands the joint as **partial** (018 earned; 019/020/021 pending) — not complete. It adds **zero** new counted knobs (an EARNED/structural fingerprint slice, like Stages 016/011/012/014);  $c_s$  is a units carrier only (the earned rationals and  $\chi_Q$  are  $c_s$ -free), cited from Stage-005 R1 as provenance, not a consumed value.

**The outgoing DtN log-derivative.** The exterior  $\ell = 2$  outgoing solution is the spherical Hankel function  $h_2^{(1)}(z) = j_2(z) + i y_2(z)$ ,  $z = a\omega/c_s$ , with  $j_2 = (3/z^3 - 1/z) \sin z - (3/z^2) \cos z$ ,  $y_2 = (1/z - 3/z^3) \cos z - (3/z^2) \sin z$  built from explicit rational  $\cdot \sin / \cos$  (no stored artifact). The DtN eigenvalue and normalized response are

$$\Lambda_2^{\text{out}}(z) = z \frac{h_2^{(1)'}(z)}{h_2^{(1)}(z)}, \quad \hat{Y}_2^{\text{out}}(z) = -\frac{3}{\Lambda_2^{\text{out}}(z)},$$

where  $-3 = -(\ell + 1)$  for  $\ell = 2$  is the outgoing static DtN slot (self-derived). This is self-contained exterior-wave algebra: it does *not* consume the interior port kernel (that enters Stage-019's prefactor).

**The fingerprint (the EARNED headline).** Expanding  $\hat{Y}_2^{\text{out}}$  about  $z = 0$ ,

$$\hat{Y}_2^{\text{out}}(z) = 1 + \frac{1}{9}z^2 + \frac{4}{81}z^4 + i\frac{1}{27}z^5 + O(z^6),$$

so  $u_2^z = 1/9$ ,  $u_4^z = 4/81$ ,  $v_5^z = 1/27$  (the odd  $z^1, z^3$  vanish; the  $z^5$  term is purely imaginary — the radiating part,  $\text{coeff}(z, 5)/i$ ). Restoring units through  $z = a\omega/c_s$ :  $u_2 = a^2/9c_s^2$ ,  $u_4 = 4a^4/81c_s^4$ ,  $v_5 = a^5/27c_s^5$ . The 27 is EARNED as  $1/v_5^z$  — the radiative-slot denominator, series-derived, *not* a typed literal (this is the `derived_in_gate` 27 that Stage-020's  $54/5 = 2 \cdot 27/5$  partition rides). The coefficients are read off the *actual* symbolic series of  $-3/(z h_2^{(1)'}/h_2^{(1)})$  with  $h_2^{(1)}/\Lambda_2^{\text{out}}/\hat{Y}_2^{\text{out}}$  emitted — not hardcoded and compared. Perturbing the typed target ( $u_2^z$  expected  $1/9 \rightarrow 1/8$ ) fires the assert, and mutating the *derivation* ( $h \cdot (1 + z^4) \rightarrow u_4^z = 112/81$ , leaving  $u_2^z, v_5^z$ ) fires exactly that coefficient's `expect_fail` — a genuine per-coefficient firewall.

**The  $\chi_Q$  sign.** Against the canonical radiative slot  $a^5/(27c_s^5)$ ,  $\chi_Q = v_5/(a^5/27c_s^5) = +1$  outgoing ( $h_2^{(1)}$ ) and  $-1$  incoming ( $h_2^{(2)} = j_2 - i y_2$ ); the sign *emerges* from  $j_2 \pm i y_2$  (the  $z^5$  coefficient flips), never a typed  $+1$ . The standing branch  $j_2$  gives  $\Lambda_{\text{stand}}(0) = +2$  ( $= +\ell$ , not the outgoing  $-3$ ),  $\hat{Y}_{\text{stand}}(0) = -3/2 \neq 1$ , and no radiating term — proving the  $+1/27$  rational is outgoing-BC-selected, not universal.

**Passivity and the units-restored dim leg.** The radiating  $v_5\omega^5$  arises from the genuine outgoing BC: the imposed-dissipation probe replaces it with a phenomenological sink and is rejected (`FAIL_NOT_OUTGOING`) by source-label provenance, with a genuine *dynamic* self-ablation re-run (with  $\neq$  without, not a constant — the `pathA_33 v1` trip-up avoided); the static $\leftrightarrow$ dynamic consistency is the fingerprint's own static slot  $\hat{Y}_2^{\text{out}}(\omega \rightarrow 0) = 1$  (*not* the prefactor  $P_0 = N_0/D_0$ , which is 019's). The units-restored dim leg runs on the real physical coefficients ( $[u_2] = T^2$ ,  $[u_4] = T^4$ ,  $[v_5] = T^5$ , with  $[a] = L$ ,  $[c_s] = LT^{-1}$ ); corrupting an  $a/c_s$  power fires `FAIL_DIMENSIONAL`. This is *not* the  $\hat{\mu}_0$ -free  $[P_0^{\text{phys}}] = 1$  gate (that is 021's) — the 018 slice contains no  $\hat{\mu}_0/N_0/D_0/P_0/\text{build\_dimensions}$  construct, keeping the back-solved- $\hat{\mu}_0$  v1 rig out.

**Honest scope.** Earned exterior-wave signature / calibrated magnitude: 018 derives the fingerprint shape ( $1/9, 4/81, 1/27$ ) and the  $\chi_Q = +1$  sign, but the  $54/5$  normalization magnitude and  $G$  are calibrated ( $G = \text{GENUINE\_BLOCKED}$ ), landed at Stage 020; the prefactor algebra is Stage 019; the  $\hat{\mu}_0$ -free dimensional closure is Stage 021. The actual-branch  $a$ -scaling and the numerical port scalars  $N_n/D_n$  are deferred Gate 4/5/6 work. The  $\chi_Q$  reconciliation (`pathA_33 +1` vs `pathA_22b`  $\approx 0.712$  — same name, different computations) is a tracked Part-VII item, not merged here.

**Consumed closure and register.** *Provenance only, no dual-site:* 018 is self-contained exterior-wave algebra, so — unlike Stage-017's genuine cross-stage dual-site — there is no checkable relation to guard. 017's  $\ell = 2$  port kernel (the interior wall mode; literal  $N_n/D_n = 019$ 's), 009/010's bulk Helmholtz mode, and  $c_s$  (Stage-005 R1) are cited as narrative provenance with no guard on any object the fingerprint never uses;  $c_s$  is an inert units carrier, and  $c_S$  (the frozen-wall speed) is not a live symbol here. Register: **zero** new counted knobs; structural edge **R37** (the outgoing-DtN  $\ell = 2$  Hankel-fingerprint provenance —  $u_2 = a^2/9c_s^2$ ,  $u_4 = 4a^4/81c_s^4$ ,  $v_5 = a^5/27c_s^5$  with the 27 computed from  $v_5^z$ , and  $\chi_Q = +1$  outgoing /  $-1$  incoming; discharges nothing). Part-II counted CALIB set unchanged at  $\{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$ .

**Verification.** Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed; the `.w1` KEEPS its native `j2/y2/Series/ Coefficient` route (its own `dim0f`),

not a mirror, agreeing at the transcript level ( $\Lambda_2^{\text{out}}/\hat{Y}_2^{\text{out}}$ , the derived  $u_2^z = 1/9/u_4^z = 4/81/v_5^z = 1/27$ ,  $\chi_Q = +1/\text{incoming } -1$ , the standing static slot, the PARTIAL landing), with an arity + unevaluated-leakage self-check. SymPy 59 PASS / Mathematica 65 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend (both engines compute-verified the series  $1 + z^2/9 + 4z^4/81 + iz^5/27$ , the incoming  $\chi_Q = -1$ , and the standing  $\Lambda_{\text{stand}}(0) = +2$ ; three non-blocking clarity nits folded — the  $P_0$  name collision, the passivity scope, the 3a/3b self-ablation scoping). Full tri-review on fresh agents: FIDELITY\_CLEAN (an independent read hand-re-derived  $j_2/y_2$ ,  $\Lambda_2^{\text{out}}$ ,  $\hat{Y}_2^{\text{out}}$ , the full series,  $\chi_Q = \pm 1$ , the standing slot) and ADVERSARIAL\_CLEAN (per-tooth ablation: 30/32 targeted mutations fired at their own assert, the 2 non-fires harness artifacts re-covered by corrected runs; no vacuous tooth) — no remediation required. Symbolic per-tooth ablation, mutations on copies.

## B.14 Stage 019: The Squared-Denominator Prefactor Algebra $P(\omega) = D_0 N / D^{\text{cons}^2}$ (Check II-G4b)

**Part / anchor.** Part II – Gravity (the frozen-throat  $\ell = 2$  radiative sector; the SECOND leg of a four-way split of pathA\_33 — the squared-denominator prefactor-algebra component of the joint QUAD\_CALIBRATED, with the DtN Hankel fingerprint = Stage 018 (done), the  $54/5 = 2 \cdot 27/5$  provenance partition + the CALIBRATED label = Stage 020, and the  $\hat{\mu}_0$ -free dimensional closure = Stage 021)

**Claim status.** EXACT WITHIN CLOSURE

**Purpose.** Earn the prefactor structure the quadrupole normalization rides. Show the squared-denominator prefactor  $P(\omega) = D_0 N(\omega) / D^{\text{cons}}(\omega)^2$ ,  $N = N_0 + N_2\omega^2 + N_4\omega^4$ ,  $D^{\text{cons}} = D_0 + D_2\omega^2 + D_4\omega^4$ , Taylor-expands about  $\omega = 0$  to  $P_0 = N_0/D_0$ ,  $P_2 = (D_0N_2 - 2D_2N_0)/D_0^2$ ,  $P_4 = (D_0^2N_4 - 2D_0(D_2N_2 + D_4N_0) + 3D_2^2N_0)/D_0^3$  — with the  $-2D_2N_0$  factor-of-2 the signature of the squared denominator (plain  $N/D$  gives only  $-D_2N_0$ , provably wrong). Land the joint verdict as a partial component.

**Status.** EXACT WITHIN CLOSURE — the prefactor algebra is exact symbolic: the coefficients are exact rational functions of the abstract port scalars, read off a symbolic **series**(...) of  $D_0N/D^{\text{cons}^2}$  (**expect\_zero**/ **expect\_bool**/ **expect\_fail** residuals, no **scipy**/**numpy**/ floats; the light integer sample substitution is a transcript-level cross-check only). The verdict is the joint QUAD\_CALIBRATED; this stage carries its *squared-denominator prefactor-algebra* component (2/4) and lands the joint as **partial** (018 done, 019 earned here; 020/021 pending) — not complete. It adds **zero** new counted knobs (an EARNED/structural prefactor-algebra slice, like Stages 018/016/011/012/014); the abstract  $D_0, D_2, D_4$  are Stage-017's exported D-lanes and the  $N_0, N_2, N_4$  are the deferred port  $N$ -moments, cited as provenance, not consumed values.

**The squared-denominator prefactor object.** The port numerator and the conservative (“consumption”) denominator are

$$N(\omega) = N_0 + N_2\omega^2 + N_4\omega^4, \quad D^{\text{cons}}(\omega) = D_0 + D_2\omega^2 + D_4\omega^4,$$

and the prefactor is the *squared-denominator* object

$$P(\omega) = \frac{D_0 N(\omega)}{D^{\text{cons}}(\omega)^2} \quad (\text{not the plain } N/D).$$

The  $D_n$  are Stage-017's exported  $\ell = 2$  port-kernel D-lanes and the  $N_n$  are the concrete port  $N$ -moments of **build\_port\_moments**; here they are carried as abstract, port-agnostic symbols — the algebra below

holds for *any* nonzero  $D_0 \dots N_4$ , so no port value is consumed (the numerical  $(D_n, N_n)$  are deferred Gate-6 branch data).

**The prefactor algebra (the EARNED headline).** Expanding  $P(\omega)$  about  $\omega = 0$  and reading the coefficients off the actual series,

$$P_0 = \frac{N_0}{D_0}, \quad P_2 = \frac{D_0 N_2 - 2D_2 N_0}{D_0^2}, \quad P_4 = \frac{D_0^2 N_4 - 2D_0(D_2 N_2 + D_4 N_0) + 3D_2^2 N_0}{D_0^3}.$$

The  $-2D_2 N_0$  term is the *signature* of the squared denominator: it arises only from the  $-2x$  term of  $(1+x)^{-2}$  with  $x = (D_2/D_0)\omega^2 + (D_4/D_0)\omega^4$ . The coefficients are  $\text{coeff}(\omega, n)$  off the *actual* symbolic series of  $D_0 N/D^{\text{cons}^2}$  — with the object and its series emitted — and are checked against an *independent* typed reference (a genuine derive-then-check that could disagree), not hardcoded and compared. Mutating the object (e.g. to plain  $N/D$ , or perturbing a  $D_n/N_n$  power) changes the extracted coefficient and fires exactly that coefficient's `expect_fail`; a swapped-in *correct* object flips the discriminator to true — a genuine positive control.

**The  $N/D$  self-check.** The plain (non-squared) object exposes the factor of two directly:  $N/D^{\text{cons}}$  series-expands to  $P_2^{\text{plain}} = (D_0 N_2 - D_2 N_0)/D_0^2$ , so the computed gap

$$P_2^{\text{plain}} - P_2 = \frac{D_2 N_0}{D_0^2} \neq 0,$$

i.e.  $-D_2 N_0$  versus the correct  $-2D_2 N_0$ . The gap is *computed* (not asserted); the counterfactual probe `3g_wrong_prefactor_object` replaces  $D_0 N/D^{\text{cons}^2}$  with plain  $N/D$ , so `plain_equals_correct_P2 = False` and the probe fires `FAIL_PREFACTOR_ALGEBRA`, while the correct object does *not* fire. Its self-ablation is a genuine *dynamic* re-run of an 019-local verdict gate (the prefactor match  $\wedge$  the  $N/D$  self-check; with  $\neq$  without, never a constant, and never the joint `base_verdict`) — the pathA\_33 v1 trip-up avoided.

**Provenance-only consumption and units-free scope.** This is where Stage-018's deferred literal  $N_n/D_n$  port-kernel consumption lands — at the *provenance* level, not a value-consumption dual-site. Because the algebra is port-agnostic and `build_port_moments` is emitted-but-never-checked (deferred Gate-6 branch data), there is no checkable cross-stage relation to guard: Stage-017's D-lanes, the concrete  $N$ -moments, and Stage-018's fingerprint context are cited as narrative provenance with no guard/dual-site on any of them (the same provenance-only landing as Stage 018; a guard on the unused/deferred moments would be a vacuous tooth). The stage is *units-free*: it introduces no  $c_s/a/G/\hat{\mu}_0$  and has no dimensional leg (the  $\hat{\mu}_0$ -free  $[P_0^{\text{phys}}] = 1$  closure is Stage-021's). A runtime free-symbol guard asserts the earned expressions' symbols are exactly  $\{\omega, D_0, D_2, D_4, N_0, N_2, N_4\}$ .

**Honest scope.** Earned prefactor structure / calibrated magnitude: 019 derives the port-agnostic algebra ( $P_0, P_2, P_4$  and the  $N/D$  factor-of-two self-check), but the  $54/5 = 2 \cdot 27/5$  normalization magnitude and  $G$  are calibrated ( $G = \text{GENUINE\_BLOCKED}$ ), landed at Stage 020, and the numerical port scalars  $(D_n, N_n)$  from the solved nonlinear branch remain deferred Gate-6 work. The DtN fingerprint is Stage 018; the  $\hat{\mu}_0$ -free dimensional closure is Stage 021.

**Consumed closure and register.** *Provenance only, no dual-site:* the abstract  $D_0, D_2, D_4$  are Stage-017's exported D-lanes and the  $N_0, N_2, N_4$  are **build\_port\_moments**' concrete port  $N$ -moments (deferred Gate-6, asserted against nothing); Stage-018's fingerprint context is a narrative cite. Register: **zero** new counted knobs; structural edge **R38** (the squared-denominator prefactor-algebra provenance —  $P_0 = N_0/D_0$ ,  $P_2 = (D_0N_2 - 2D_2N_0)/D_0^2$ ,  $P_4 = (D_0^2N_4 - 2D_0(D_2N_2 + D_4N_0) + 3D_2^2N_0)/D_0^3$  with the  $-2D_2N_0$  factor-of-2 the squared-denominator signature; discharges nothing). Part-II counted CALIB set unchanged at  $\{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$ .

**Verification.** Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed; the .wl KEEPS its native **serW/Coefficient/ FullSimplify** route on its own  $P(\omega)$ /plain objects (typing its own expected  $P_0/P_2/P_4$ ), not a mirror, agreeing at the transcript level (the series-extracted coefficients, the plain  $P_2 = (D_0N_2 - D_2N_0)/D_0^2$ , and **plainEqualsCorrectP2=False**), with an arity + unevaluated-leakage self-check. SymPy 18 PASS / Mathematica 24 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend (both engines compute-verified the series  $N_0/D_0 + (D_0N_2 - 2D_2N_0)\omega^2/D_0^2 + \dots$ , the plain  $N/D$  gap  $D_2N_0/D_0^2$ , and the emitted-but-never-checked reading of **build\_port\_moments**; four nits folded — a line-ref fix, the D-lane/ $N$ -moment split, the sibling-symbol wording, and the  $D_0..N_4$ -only sample substitutions). Full tri-review on fresh agents: **FIDELITY\_CLEAN** (an independent read hand-re-derived  $P_0/P_2/P_4$  and the  $N/D$  gap  $D_2N_0/D_0^2$ ) and **ADVERSARIAL\_ISSUES** (per-tooth ablation: the central firewall,  $N/D$  self-check, and dynamic 019-local ablation all genuine; two subsumed/mirror teeth flagged — one made-genuine as a swapped-in correct-object positive control, one honestly de-counted — plus a de-obfuscation of narrative tokens and a legibility fix, all re-verified **REVERIFY\_CLEAN** by the coupling meta-test). Symbolic per-tooth ablation, mutations on copies.

## B.15 Stage 020: The $54/5 = 2 \cdot 27/5$ Provenance Partition and the CALIBRATED Verdict Label (Check II-G4c)

**Part / anchor.** Part II – Gravity (the frozen-throat  $\ell = 2$  radiative sector; the THIRD leg of a four-way split of pathA\_33 — the  $54/5 = 2 \cdot 27/5$  provenance-partition + the CALIBRATED-verdict-label component of the joint **QUAD\_CALIBRATED**, with the DtN Hankel fingerprint = Stage 018 (done), the squared-denominator prefactor algebra = Stage 019 (done), and the  $\hat{\mu}_0$ -free dimensional closure = Stage 021 (the completing leg))

**Claim status.** EXACT WITHIN CLOSURE

**Purpose.** Land the CALIBRATED headline. Show the assembled  $\ell = 2$  quadrupole magnitude  $54Gc_s^5/(5a^5c^5)$  decomposes as  $54/5 = 2 \cdot 27/5$ , with the 27 earned (**derived\_in\_gate**, Stage-018's fingerprint  $1/v_5^z$ ) and the  $2/5 + G$  calibrated (**external\_bridge\_input**, the GR Burke–Thorne  $2G/5c^5$ ,  $G = \text{GENUINE_BLOCKED}$ ); a four-way provenance partition classifies the assembled  $54/5$  as **external\_bridge\_input** (external dominates), so the source-faithful verdict lands **QUAD\_CALIBRATED** (not **QUAD\_PASS**) — driven by provenance, not by  $G \rightarrow \lambda G$  invariance.

**Status.** EXACT WITHIN CLOSURE — units-bearing but exact symbolic/rational for the earned content: the  $a^{-5}$  scaling (exact  $a$ -power via Stage-018's frozen slot), the  $\Gamma_5/\chi_Q$  equivalence (**compact(...)** = 0 residuals), the  $54/5 = 2 \cdot 27/5$  SymPy rational identity (bound to **target\_rhs/v5\_slot**), and the four-way provenance partition (exact class comparison via tag dominance); **expect\_zero/expect\_bool/**

`expect_fail` residuals, no floats/tolerances in the earned content. The verdict is the joint `QUAD_CALIBRATED`; this stage carries its *provenance-partition* + *CALIBRATED-label* component (3/4) and lands the joint as **partial** (018/019 done, 020 earned here; 021 the completing leg) — and it is the leg where the *CALIBRATED classification* is determined. It adds **zero** new counted knobs (a classification slice, like Stages 018/019).

**The assembled magnitude and the  $\Gamma_5/\chi_Q$  equivalence.** The  $\ell = 2$  quadrupole normalization the gravity sector must match is the Burke–Thorne form

$$\hat{m}_0^2 P_0 = \frac{54 G c_s^5}{5 a^5 c^5} \iff \gamma_{\text{quad}}^{\text{eff}} = \frac{2G}{5c^5}.$$

The equivalence is a self-contained algebraic bridge. With  $P_0^{\text{phys}} = (c_s/a)^2 (N_0/D_0)$  (Stage-019’s prefactor) and Stage-018’s outgoing  $\chi_Q = +1$ , the forward residual

$$\text{forward} = \frac{54 G c_s^5}{5 a^5 c^5} \cdot \chi_Q \cdot \frac{a^5}{27 c_s^5} - \frac{2G}{5c^5} = \frac{2G(\chi_Q - 1)}{5c^5}$$

vanishes *iff*  $\chi_Q = 1$  and  $54/27 = 2$  (the 27 is Stage-018’s radiative slot  $1/v_5^z$ ). So the outgoing sign Stage 018 derived is exactly the sign a positive quadrupole demands; a wrong  $2/5 \rightarrow 3/5$  breaks the bridge (3e→FAIL\_EQUIVALENCE).

**The  $54/5 = 2 \cdot 27/5$  decomposition (the headline — a bound SymPy identity).** The assembled coefficient factors as

$$\frac{54}{5} = 2 \cdot \frac{27}{5},$$

with the 27 = `derived_in_gate` (Stage-018’s fingerprint  $1/v_5^z$ ) and the  $2/5+G$  = `external_bridge_input` (the GR Burke–Thorne  $2G/5c^5$ ). This is a SymPy-verified rational identity *bound to the stage’s own assembled expressions* — not the source’s typed string “ $54/5 = 2 \cdot 27/5$ ”, and not a bare-literal `Rational(54,5) - 2 * Integer(27)/5` arithmetic tautology. The magnitudes are extracted: `mag = compact(target_rhs/(G*c_s^5/(a^5*c^5))) = 54/5` and `27_slot = compact(a^5/(c_s^5*v5_slot)) = 27` (from Stage-018’s frozen slot `v5_slot = a^5/27c_s^5`), then `compact(mag - 2*27_slot/5) = 0`. Mutating `target_rhs` (54 → 55) or the identity ( $2 \cdot 26/5$ ) fires the assert.

**The  $a^{-5}$  target scaling.** The assembled magnitude carries  $a^{-5}$  as a *derived* consequence of  $a$ -cancellation, not a typed target power: the Burke–Thorne coefficient  $2G/5c^5$  is  $a$ -free, and Stage-018’s radiative slot `v5_slot = a^5/27c_s^5` supplies the  $a^5$ , so `derived_power = a-power( $\gamma_{\text{target}}$ ) - a-power(v5_slot) = 0 - 5 = -5`, and the reconstructed  $\gamma_{\text{target}}/(\chi \cdot \text{v5\_slot}) = \text{target\_rhs}$  ( $a$ -power  $-5$ ). The  $a^5$  is typed *once*; the wrong-scaling probe ( $a^5 \rightarrow a^4$ ) fires at the assembled residual (3c→FAIL\_SCALING). The actual branch  $a$ -scaling (from  $N_0/D_0$ ) is deferred Gate-6.

**The four-way provenance partition and the PROVENANCE-driven verdict.** Every quantity is classified from its provenance *tags* by a tag-dominance rule `deferred > external > derived > convention`: `fingerprint_27` → `derived_in_gate`, `PN_2_over_5` → `external_bridge_input`,  $G$  → `external_bridge_input`, and the assembled  $54/5$  (mixing the earned 27 tag with the external  $2/5$  tag) → `external_bridge_input` because *external dominates*. The 020-local verdict then reads the classes with the source-faithful rule

```
if g_class = derived & mag_class = derived: QUAD_PASS; else: QUAD_CALIBRATED,
```

so with  $G$  and the assembled 54/5 both external, the verdict lands **QUAD\_CALIBRATED**: the PDE delivers the form/branch (the 27), not Newton's  $G$  ( $G = \text{GENUINE\_BLOCKED}$ ). The classifier's dominance rule is proven by a truth-table + key-class + tag-mutation tooth (the source **partition\_ok** alone is trivially true), and the verdict is proven to *read* provenance by a **QUAD\_PASS** positive control (a coherent all-derived partition) *and* a required mixed control (one external, one derived  $\rightarrow$  **QUAD\_CALIBRATED**, the case an inverted rule would mislabel).

**The verdict is provenance-driven, not  $G \rightarrow \lambda G$ -invariance-driven.** A separate, non-verdict-driving diagnostic makes the point sharp: 54/5 is a pure number, hence  $G$ -invariant ( $G \rightarrow \lambda_G G$  leaves it unchanged), yet it is **external\_bridge\_input** by provenance. An invariance-only test would *mislabel* the calibrated 54/5 as earned (**invariance\_only\_trap\_catches\_54\_over\_5** = True); the classification is set by the provenance tags, and **3f\_partition\_mislabel** fires **FAIL\_PROVENANCE\_PARTITION** in both directions ( $G$  external $\rightarrow$ derived and 27 derived $\rightarrow$ external).

**Units-bearing scope (no dimensional-homogeneity gate).** Unlike Stage 019 (units-free), Stage 020 is units-bearing —  $\{c_s, a, c, G\}$  live in the scaling/equivalence algebra — but it does algebra + provenance, *not* the dimensional homogeneity gate  $[P_0^{\text{phys}}] = 1$  (that is Stage-021's). The cut is by operation, enforced by a runtime free-symbol allowlist (no  $\hat{\mu}_0$  symbol; the source's legacy **gamma\_quad\_eff**  $\hat{\mu}_0$  string is dropped), and a structural no-dimensional-dependency cut (no **dim\_of** construct; a structural assertion that the 020-local verdict's signature/bytecode carries no dim parameter/dependency — the **.wl** checks its definition arity/text).

**Honest scope.** Earned classification / calibrated magnitude: 020 earns the  $a^{-5}$  scaling, the  $\Gamma_5/\chi_Q$  equivalence, the 54/5 = 2·27/5 decomposition, and the four-way provenance partition; the 54/5 magnitude and  $G$  are calibrated ( $G = \text{GENUINE\_BLOCKED}$ ). The  $c^5$  is the  $\text{GR}/\lambda_\gamma$  units bridge ( $P_0 \propto c_s^5/c^5 = 1/\lambda_\gamma^5$ ) —  $c$  is the light cone  $c_\gamma$  in its GR-bridge role, a benchmark carrier, not EM propagation and not a fresh knob. The actual branch  $a$ -scaling and the numerical  $(D_n, N_n)$  port scalars remain deferred Gate-6. The dimensional closure  $[P_0^{\text{phys}}] = 1$  is Stage-021's.

**Consumed closure and register.** *Provenance, no dual-site:* Stage-018's  $\chi_Q = +1$  and 27 ( $= 1/v_5^2$ ) genuinely enter 020's self-contained equivalence bridge (a corrupt cited value breaks **forward** — the equivalence tooth itself, not a manufactured cross-stage guard); Stage-019's  $P_0 = N_0/D_0$  appears only in the  $\Gamma_5$  definition, absent from the equivalence ok residual (a  $\{N_0, D_0\} \subseteq \Gamma_5 \wedge \notin$  residual firewall guards this); Stage-017's D-lanes are the provenance of the abstract port scalars. Register: **zero** new counted knobs; structural edge **R39** (the 54/5 = 2·27/5 provenance-partition + the **CALIBRATED**-verdict provenance; discharges nothing — a classification landing, not a reduction).  $G = \text{GENUINE\_BLOCKED}$ /external (already registered);  $c$  = the  $c_\gamma$  GR-units-bridge (cited benchmark); the 2/5 = GR bridge; the 27 = Stage-018's derived fingerprint. Part-II counted **CALIB** set unchanged at  $\{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$ .

**Verification.** Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed. Unlike Stages 018/019 (which kept the **.wl**'s already-native blocks), the source **.wl** had *zero* 020 content, so the 020 **.wl** is a genuinely fresh independent authoring: native **Exponent/Together/Cancel/FullSimplify** for the scaling + the  $\Gamma_5$  bridge, native **Rational/Simplify** for the identity bound to its own **targetRHS/ v5Slot**, a native **Association**-based provenance classifier using a rank-based **MaximalBy** (a genuinely different mechanism from the **.py** if-chain, same dominance), and **expr/.G**  $\rightarrow \lambda_G G$  for the  $g$ -invariance trap — no **Series**, no **.py** mirroring — agreeing at the



transcript level (the  $a^{-5}$ , **forward** = 0, the bound  $54/5 = 2 \cdot 27/5$ , the provenance classes, the CALIBRATED verdict with the QUAD\_PASS and mixed controls, both 3f directions), with an arity + unevaluated-leakage self-check. SymPy 74 PASS / Mathematica 82 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend: Codex folded four blocking genuineness gaps; Grok's compute-verify then caught the CALIBRATED verdict-rule *inversion* (the source default is **else**→QUAD\_CALIBRATED; the directive had inverted it to **PASS** unless both external — a shippable rig, fixed by a required mixed control) plus a bare-literal-identity nit; a Codex confirm then swept two consistency gaps, all folded to DIRECTIVE\_CLEAN. Full tri-review on fresh agents: FIDELITY\_CLEAN (an independent read hand-re-derived the  $\Gamma_5$  bridge, the bound identity, the classify-provenance dominance, and the g-invariance trap) and ADVERSARIAL\_ISSUES (per-tooth ablation: the four key genuineness risks — the bound identity, the mixed control catching the inverted rule, the proven classifier, the genuinely-authored .w1 — all clean; three low-severity vacuous/subsumed teeth flagged, all remediated make-genuine with no de-counts — the structural dimensional cut, the  $\{N_0, D_0\} \subseteq \Gamma_5$  before/after firewall, the independently-classified baseline-external side — re-verified REVERIFY\_CLEAN by the coupling meta-test). Symbolic per-tooth ablation, mutations on copies.

## B.16 Stage 021: The $\hat{\mu}_0$ -free $[P_0^{\text{phys}}] = 1$ Dimensional Closure (Check II-G4d)

**Part / anchor.** Part II – Gravity (the frozen-throat  $\ell = 2$  radiative sector; the FOURTH, completing leg of a four-way split of pathA\_33 — the  $\hat{\mu}_0$ -free  $[P_0^{\text{phys}}] = 1$  dimensional-closure component of the joint QUAD\_CALIBRATED, with the DtN Hankel fingerprint = Stage 018 (done), the squared-denominator prefactor algebra = Stage 019 (done), and the  $54/5 = 2 \cdot 27/5$  provenance partition + the CALIBRATED verdict label = Stage 020 (done))

**Claim status.** EXACT WITHIN CLOSURE

**Purpose.** Complete the joint QUAD\_CALIBRATED. Certify the assembled  $\ell = 2$  quadrupole prefactor  $P_0^{\text{phys}} = (c_s/a)^2(N_0/D_0)$  is dimensionless from the sourced port dimensions  $[N_0] = L^{-1}M$ ,  $[D_0] = L^{-1}T^{-2}M$  — a  $\hat{\mu}_0$ -FREE gate that catches the natural-units trap (the handoff's  $P_0 = N_0/D_0$  silently drops  $(c_s/a)^2$ ) and reads the sourced port dimensions (corrupt- $[N_0]$  fires FAIL\_DIMENSIONAL) rather than a back-solved free carrier  $\hat{\mu}_0$  (corrupt- $[G]$  does not fire).

**Status.** EXACT WITHIN CLOSURE — exact symbolic  $(L, M, T)$ -triple dimensional-vector algebra, float-free: the  $\hat{\mu}_0$ -free gate ( $\text{dim\_of}(P_0^{\text{phys}}) = \text{ZERO\_DIM}$ ), the natural-units-trap catch (dropping the frequency normalization leaves  $T^2 \neq 1$ ), the corrupt-dim scope truth-table, and the anti-v1 back-solve-tautology demotion; **expect\_zero/expect\_bool/expect\_fail** residuals, no floats/tolerances. The verdict is the joint QUAD\_CALIBRATED; this stage carries its *dimensional-closure* component (4/4) and lands the joint **complete** (018/019/020 done, 021 earned here). It adds **zero** new counted knobs (a dimensional-closure slice, like Stages 018/019/020).  $\hat{\mu}_0$  is a free-carrier non-verdict diagnostic, not a counted knob.

**The  $\hat{\mu}_0$ -free  $[P_0^{\text{phys}}] = 1$  gate.** With the sourced port dimensions  $[N_0] = L^{-1}M$  (the density/continuity-port numerator) and  $[D_0] = L^{-1}T^{-2}M$  (the carried reduced static conservative denominator  $D_0 = K - B_0 - Z_0$ ), and the frequency normalization  $(c_s/a)^2$ ,

$$[P_0^{\text{raw}}] = [N_0/D_0] = T^2, \quad [(c_s/a)^2] = T^{-2}, \quad [P_0^{\text{phys}}] = [(c_s/a)^2 N_0/D_0] = 1,$$

so `dimensional_ok = True`. The verdict-bearing check is `dimensional_ok = (dim_of( $P_0^{\text{phys}}$ ) = ZERO_DIM)` — a computed  $(L, M, T)$ -triple equality reading the sourced dims, not a typed "1". Crucially  $\hat{\mu}_0$  does *not* enter the verdict: it is absent from `raw_symbol_dims` (the gate's dimension map), and the verdict's computed read-set excludes  $\hat{\mu}_0/\mu_{\text{dim}}/\text{homogeneity\_pass}$ .

**The natural-units-trap catch (the dimensional milestone).** The handoff's  $P_0 = N_0/D_0$  silently drops the  $(c_s/a)^2$  frequency-normalization factor (a natural-units artifact where  $c_s/a$  was set to 1). The gate catches it:  $[P_0^{\text{raw}}] = T^2 \neq 1$ , so the 3d drop-normalization probe fires `FAIL_DIMENSIONAL`. The frequency normalization is required for  $[P_0^{\text{phys}}] = 1$ .

**The corrupt-dim scope truth-table.** The  $\hat{\mu}_0$ -free gate fires precisely on the dims that enter  $P_0^{\text{phys}}$ :

|                               |                                                                                          |                                              |
|-------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------|
| corrupt $[N_0] \rightarrow 0$ | $[P_0^{\text{raw}}] = (1, -1, 2)$ , $[P_0^{\text{phys}}] = (1, -1, 0) = L M^{-1} \neq 1$ | $\rightarrow \text{FAIL\_DIMENSIONAL (3d')}$ |
| corrupt $[D_0] \rightarrow 0$ | $[P_0^{\text{phys}}] = (-1, 1, -2) \neq 1$                                               | $\rightarrow \text{FAIL\_DIMENSIONAL}$       |
| corrupt $[c_s] \rightarrow 0$ | $[P_0^{\text{phys}}] = (-2, 0, 2) \neq 1$                                                | $\rightarrow \text{FAIL\_DIMENSIONAL}$       |
| corrupt $[G] \rightarrow 0$   | $[P_0^{\text{phys}}] = 1$ still ( $G \notin \text{free}$ )                               | $\rightarrow \text{NO\_FAIL (scope)}$        |
| correct                       | $[P_0^{\text{phys}}] = 1$                                                                | $\rightarrow \text{NO\_FAIL}$                |

The corrupt- $[N_0]$  result is  $[P_0^{\text{phys}}] = (1, -1, 0)$  — the  $(c_s/a)^2$  factor *remains* (it is not  $-[D_0] = (1, -1, 2)$ , which is  $[P_0^{\text{raw}}]$ ). The gate is G-independent; the corrupt- $[G]$  row is a dependency-scope diagnostic, asserted via the runtime test  $G \notin \text{free\_symbols}(P_0^{\text{phys}})$ .

**The  $\hat{\mu}_0$  homogeneity diagnostic is NON-verdict — the anti-v1 discriminator.** The full closure  $\hat{m}_0^2 P_0 = 54 G c_s^5 / (5 a^5 c^5)$  can be made dimensionally homogeneous by back-solving the free carrier  $[\hat{\mu}_0] = L^{-1} T^{-1} M^{-1/2}$  from  $(\text{rhs\_dim} - \text{p0\_dim})/2$ , giving  $[\text{lhs}] = [\text{rhs}] = L^{-2} T^{-2} M^{-1}$ , `homogeneity_pass = True`. This is a *tautology*: re-solving  $[\hat{\mu}_0]$  after *any* corruption keeps `homogeneity_pass = True` (the back-solve absorbs the change), so it can never fail — exactly the v1-rejected rig, correctly demoted here to a labelled diagnostic (`participates_in_verdict = False`, computed from the verdict read-set). The decisive anti-v1 tooth is two-part: (a) a computed read-set assertion that the verdict excludes  $\hat{\mu}_0/\mu_{\text{dim}}/\text{homogeneity\_pass}$  (and  $\hat{\mu}_0 \notin \text{raw\_symbol\_dims}$ ), and (b) a deliberately-wired mutant verdict = the back-solved `homogeneity_pass`, re-run over all four corruptions and shown to stay `NO_FAIL` on all of them — proving it is vacuous, so the real verdict must be the  $\hat{\mu}_0$ -free  $[P_0^{\text{phys}}] = \text{ZERO}$  gate (which fires on three of the four). The gate's  $[N_0]/[D_0]/[c_s]$  FAIL rows, not the  $[G]$  row, are what reject the all-`NO_FAIL` back-solved gate.

**The  $\hat{Y}$  dimensionless check.** The outgoing physical expansion  $\hat{Y} = 1 + u_2 \omega^2 + u_4 \omega^4 + i v_5 \omega^5$  with Stage-018's frozen fingerprint  $u_2 = a^2/9c_s^2$ ,  $u_4 = 4a^4/81c_s^4$ ,  $v_5 = a^5/27c_s^5$  (a provenance-frozen local fixture, not the Stage-018 machinery) is dimensionless (each term  $T^2 \cdot T^{-2} = 1$ , etc.), so  $[\hat{Y}] = 1$ . The check is wrapped in a structured `try_dim_of/Catch` so a corrupted  $\omega$ -power ( $u_2 \omega^3$ ) fires the named  $\hat{Y}$ -dimensionless assert (a caught mismatch), not an uncaught `DimError` that would bypass the tooth.

**The completing landing.** Stage 021 lands the joint `QUAD_CALIBRATED complete` (018 fingerprint  $\wedge$  019 prefactor  $\wedge$  020 provenance-partition + `CALIBRATED`  $\wedge$  021 dim closure). **Complete  $\neq$  pass:** the joint token stays `QUAD_CALIBRATED` (calibrated, not `QUAD_PASS`) — Stage 020 already determined the `CALIBRATED` classification and  $G = \text{GENUINE\_BLOCKED}$ ; 021 supplies the dimensional closure that makes the assembled magnitude dimensionally sound, but does not flip the verdict. The PDE delivers the form/branch (the 27 fingerprint + the dimensionally-sound closure), not Newton's  $G$ .

**Honest scope.** Earned closure / calibrated magnitude: 021 earns the  $\hat{\mu}_0$ -free  $[P_0^{\text{phys}}] = 1$  closure, the natural-units-trap catch, and the corrupt- $[N_0]/[G]$  discriminator (the gate reads the sourced port dims, not a back-solved  $\hat{\mu}_0$ ); the 54/5 magnitude and  $G$  are calibrated ( $G = \text{GENUINE\_BLOCKED}$ ). 021 is the other half of the 020/021 operation-level cut — Stage 020 did algebra + provenance, Stage 021 does the  $[\cdot]$  dimensional-homogeneity gate, the only pathA\_33 leg with a dimensional gate. The sourced port dims  $[N_0] = L^{-1}M$  (pathA\_43 density-port numerator) and  $[D_0] = L^{-1}T^{-2}M$  (the carried reduced static conservative denominator) are sourced dimensional inputs (Cluster C stages 024/027 + pathA\_34 not yet built in build order); they genuinely enter the gate (so the corrupt- $[N_0]$  tooth is genuine), but 021 does not re-derive them. The actual branch  $a$ -scaling and the numerical  $(D_n, N_n)$  port scalars remain deferred Gate-6.

**Consumed closure and register.** *Provenance, no dual-site:* the sourced port dims (pathA\_43 numerator + the carried conservative denominator), Stage-018's  $u_2/u_4/v_5$  (a local frozen fixture for the  $\hat{Y}$  check, not `build_fingerprint`), Stage-019's  $P_0 = N_0/D_0$  (entering  $P_0^{\text{raw}}$ ), and Stage-020's assembled `target_rhs` (the  $\hat{\mu}_0$  diagnostic's rhs only). The sourced port dims genuinely enter the  $\hat{\mu}_0$ -free gate (the corrupt- $[N_0]$  probe is the genuine able-to-fail tooth). Register: **zero** new counted knobs; structural edge **R40** (the  $\hat{\mu}_0$ -free  $[P_0^{\text{phys}}] = 1$  dimensional closure; discharges nothing — a dimensional-consistency closure, not a reduction; records that 021 completes the joint `QUAD_CALIBRATED`, which stays `CALIBRATED` not `PASS`).  $\hat{\mu}_0$  is a free-carrier non-verdict diagnostic; the sourced port dims are dimensional provenance;  $c/G$  are already registered. Part-II counted `CALIB` set unchanged at  $\{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$ .

**Verification.** Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed. Like Stages 018/019 (and unlike Stage 020's fresh authoring), the source `.wl` has the 021 dimensional block, so the 021 `.wl` keeps its native, already-independent Wolfram dimensional-vector engine (a `Which`-based `dimOf` over `Times/Power/Plus`, native  $\langle |\dots| \rangle$  `rawDims`, `Series`-free dimension algebra) and severs only the YAML machinery; the Stage-018 fingerprint ( $j_2/y_2$ ) and Stage-019 prefactor blocks are removed and the  $\hat{Y}$  coefficients become local `u2Sourced/u4Sourced/v5Sourced` fixtures. It adds (the source `.wl` lacked both) the corrupt- $[G]$ -does-not-fire scope control and the dynamic 021-local self-ablation; the `.wl`'s local pass-guard uses a split-out `yhatOk` and excludes `homogeneityPass` (a print-only diagnostic), with an arity + unevaluated-leakage self-check. Agreement is at the transcript level ( $[P_0^{\text{raw}}] = T^2$ ,  $[(c_s/a)^2] = T^{-2}$ ,  $[P_0^{\text{phys}}] = 1$ , `3d/3d'` `FAIL_DIMENSIONAL`, corrupt- $[G]$  `NO_FAIL`, `homogeneity_pass = True` diagnostic). SymPy 42 `PASS` / Mathematica 50 `PASS`, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend: Codex folded six blocking findings — most notably that the originally-proposed corrupt- $[G]$  anti-v1 discriminator was *backwards* (a back-solved  $\hat{\mu}_0$  gate re-solves  $\mu_{\text{dim}}$  after every mutation, so it fires on nothing, and it is the  $\hat{\mu}_0$ -free gate's  $[N_0]/[D_0]/[c_s]$  `FAIL` rows that reject it) — then a Grok compute-verify pass independently confirmed the  $\hat{\mu}_0$ -free gate dims, the corrupt-dim truth-table, the back-solve-is-a-tautology crux, and the  $\hat{Y}$ /structured-catch, and a final Codex confirm closed it to `DIRECTIVE_CLEAN`. Full tri-review on fresh agents: `FIDELITY_CLEAN` (an independent read hand-re-derived the  $\hat{\mu}_0$ -free gate dims, the corrupt-dim truth-table, and the back-solve-tautology table, confirming no 018/019/020 leakage and a genuine independent `.wl`) and `ADVERSARIAL_ISSUES` (per-tooth ablation: the two key teeth — the anti-v1 read-set/wired-mutant and the corrupt- $[G]$  scope control — genuine; five low-severity stamped/subsumed teeth flagged, remediated two make-genuine (`rerun_gate_logic` derived from the actual re-run, `participates_in_verdict` derived from the computed read-set) and three de-count (the `QUAD`-landing tautology and two subsumed summaries retained as labeled prints) — re-verified `REVERIFY_CLEAN` by the coupling meta-test, tallies 45/53→42/50 from the honest de-counts). Symbolic

per-tooth ablation, mutations on copies.

## B.17 Stage 022: The Cross- $\ell - (\ell + 1)/\Lambda_\ell$ Fingerprints + the Gate-4 Non-Regression (Check II-G5a)

**Part / anchor.** Part II – Gravity (the frozen-throat cross- $\ell$  radiative sector; the EARNED-FIRST leg of a two-way split of pathA\_34 — the cross- $\ell$  outgoing DtN  $-(\ell+1)/\Lambda_\ell$  fingerprints ( $\ell = 0, 1, 2$ ) + the Gate-4 quadrupole non-regression component (1/2) of the joint FAIL\_UNDERDETERMINED\_NOT\_PREDICTIVE). The native nullspace underdetermination departure — dim-8 / return-nullity-2, the residuals vs pathA\_29, the  $Z_0^{\text{ret}}/Z_1^{\text{ret}}$  Gate-6 selector need — that *delivers* the FAIL is Stage 023 (II-G5b).

**Claim status.** EXACT WITHIN CLOSURE

**Purpose.** Generalize Stage 018's  $\ell = 2$ -only DtN Hankel fingerprint across  $\ell$ : certify that the outgoing DtN responses  $\hat{Y}_\ell^{\text{out}} = -(\ell + 1)/\Lambda_\ell$  for  $\ell = 0, 1, 2$  — computed from the spherical Hankel  $h_\ell$  — series-expand to the radiative fingerprint  $\{1, 1/2, 1/27\}$  at leading orders  $\{\omega^1, \omega^3, \omega^5\}$ , with the static-DtN slot  $\Lambda_\ell(z \rightarrow 0) = -(\ell + 1)$  derived from the Hankel logarithmic derivative and the radiative order verified by scanning; and that the  $\ell = 2$  leg reproduces the completed pathA\_33 quadrupole fingerprint  $\{u_2, u_4, v_5\} = \{1/9, 4/81, 1/27\}$  (the Gate-4 non-regression — the cross- $\ell$  generalization does not regress the earned quadrupole).

**Status.** EXACT WITHIN CLOSURE — exact symbolic, float-free, dimensionless  $z$ -space ( $z = a\omega/c_s$ ): the cross- $\ell$  spherical-Hankel logarithmic derivatives and their small-frequency series are exact `sympy.series` / native `Series` expansions; `bool_zero/expect_bool/expect_fail` residuals, no floats/tolerances, no units-restoration leg. The verdict is the joint FAIL\_UNDERDETERMINED\_NOT\_PREDICTIVE; this stage carries its *fingerprint + non-regression* component (1/2) and lands it as an **EARNED-FIRST PARTIAL** (022 earned; 023 pending and it is 023 that delivers the FAIL). It adds **zero** new counted knobs (an earned exterior-wave signature slice, like Stages 016/018).  $c_s$  is an R1-derived units carrier, not a counted knob; the  $\ell = 0/1$  stiffnesses and the return admittances  $Z_0^{\text{ret}}/Z_1^{\text{ret}}$  are Stage 023's, not counted here.

**Two distinct verdicts, both printed.** The script emits an *022-local* exit-0 gate and, separately, the joint physics label:

```
LOCAL_AUDIT_VERDICT: CROSS_L_FINGERPRINT_OK (fingerprints + non-regression; the exit-0 gate)
JOINT_LANDING_LABEL (PARTIAL; FAIL not evaluated by 022): FAIL_UNDERDETERMINED_NOT_PREDICTIVE
```

022's script passes (exit 0) — the earned content is correct and every tooth fires; the FAIL token is the joint gate's physics label carried as a 1/2 partial. 022 is the earned half of a gate that ultimately fails (cf. Stage 003 = earned transverse photons + the characterized FAIL\_CAUCHY departure). The 022-local verdict's computed read-set is exactly `{cross_l_fingerprints, ell2_non_regression}` and provably excludes any nullspace / return / `base_verdict` object (none is built here) — so the default-verdict-is-PASS trip-up is avoided: 022 does not compute or print a FAIL from any rigged mechanism.

**The cross- $\ell$  outgoing DtN fingerprints (the earned headline).** For each  $\ell$  the exterior outgoing solution is the spherical Hankel function of the first kind  $h_\ell^{(1)} = j_\ell + i y_\ell$  ( $z = a\omega/c_s$ ); the normalized

radiative response is  $\hat{Y}_\ell^{\text{out}}(z) = -(\ell + 1)/\Lambda_\ell(z)$  with  $\Lambda_\ell(z) = z h_\ell^{(1)'} / h_\ell^{(1)}$ . Series-expanding about  $z = 0$ ,

$$\begin{aligned} \ell = 0 : \quad \hat{Y}_0 &= 1 + i z + \dots & \Rightarrow \text{radiative\_coeff} &= 1 \text{ at order } \omega^1, \\ \ell = 1 : \quad \hat{Y}_1 &= 1 + \frac{1}{2}z^2 + i\frac{1}{2}z^3 + \dots & \Rightarrow \text{radiative\_coeff} &= 1/2 \text{ at order } \omega^3, \\ \ell = 2 : \quad \hat{Y}_2 &= 1 + \frac{1}{9}z^2 + \frac{4}{81}z^4 + i\frac{1}{27}z^5 + \dots & \Rightarrow \text{radiative\_coeff} &= 1/27 \text{ at order } \omega^5. \end{aligned}$$

The leading radiative coefficient is  $\{1, 1/2, 1/27\}$  at order  $2\ell + 1 = \{\omega^1, \omega^3, \omega^5\}$ , extracted as the imaginary coefficient  $\text{coeff}(z, 2\ell + 1)/i$  and checked derive-vs-typed — the derived side is the actual series of  $-(\ell + 1)/(z h'/h)$ , emitted, not hardcoded. The static-DtN slot  $\Lambda_\ell(z \rightarrow 0) = -(\ell + 1)$  is *derived* from the Hankel logarithmic derivative ( $h_\ell \sim z^{-(\ell+1)}$  as  $z \rightarrow 0$ ), read off `lam_series.coeff(z, 0)` — not the hand-set numerator (the source’s set-then-compare is de-rigged). The radiative order is verified by *scanning* the imaginary series for its first nonzero power and asserting all lower imaginary coefficients vanish (not the source’s dodgeable “preselected  $2\ell + 1$  coefficient is nonzero”: a  $+iz$  corruption of the  $\ell = 2$  series leaves  $z^5$  nonzero yet the true leading imaginary order becomes 1; the scan catches it). The static value  $\hat{Y}_\ell(0) = 1$  is a de-counted printed diagnostic ( $= -(\ell + 1)/\Lambda_{\text{static}}$ , subsumed by the derived  $\Lambda_{\text{static}}$ ).

**The incoming-sign-flip (the genuine sign tooth).** The incoming branch  $h_\ell^{(2)} = j_\ell - i y_\ell$  flips *only* the radiative (imaginary) sign — `radiative_coeff`  $\rightarrow \{-1, -1/2, -1/27\}$  with every lower real coefficient unchanged (`bool_zero(incoming + outgoing) = 0`). This computed sign-flip is the genuine sign content. The  $\chi_Q = \text{radiative\_coeff}/(\text{canonical slot}) = +1$  equality is a de-counted printed diagnostic — algebraically subsumed by the  $\ell = 2$  non-regression ( $27v_5 - 1 \equiv 27(v_5 - 1/27)$ ), so once the  $\ell = 2$  slot passes  $\chi_Q = 1$  is automatic; not an independent tooth.

**The Gate-4 quadrupole non-regression.** The  $\ell = 2$  leg  $(-(\ell + 1)/\Lambda_\ell|_{\ell=2} = -3/\Lambda_2)$  reproduces Stage 018’s earned quadrupole fingerprint  $u_2 = 1/9$ ,  $u_4 = 4/81$ ,  $v_5 = 1/27$ . This is a derive-vs-typed non-regression: 022 re-derives the  $\ell = 2$  tuple from its own Hankel and checks it against Stage 018’s independently-earned typed literals (not a same-machinery reconstruction of the typed side — no subsumed  $X \equiv X$ ). The cross- $\ell$   $-(\ell + 1)$  normalization gives the right  $-3$  at  $\ell = 2$ , so the general machinery does not regress the quadrupole. Each of  $u_2, u_4, v_5$  has its own coefficient-isolating mutant (the 3e incoming mutation flips only  $v_5$ , so  $u_2/u_4$  need their own derivation-copy mutants — the Stage 018 pattern); breaking the  $\ell = 2$  branch to incoming fires `FAIL_QUAD_REGRESSION`, whose self-ablation is a dynamic two-verdict re-run (with  $\neq$  without) re-scoped to the 022-local verdict.

**Honest scope.** Earned cross- $\ell$  signature + non-regression / the FAIL is 023’s: 022 derives the outgoing cross- $\ell$  fingerprints and the Gate-4 quadrupole non-regression — the report’s earned fingerprint / raw-order / Gate-4 subset. The rest of the report’s earned content (the residual form/sign/order, conditional on a positive bounded branch) plus the FAIL-delivering native nullspace underdetermination (dim-8 / return-nullity-2,  $\epsilon_{\text{eff}}$  left free at the linear Gate-5 level) and the deferred scalar/dipole magnitude/nonzero prediction belong to Stage 023. The joint FAIL is deferred to 023, not evaluated here (the `JOINT_LANDING_LABEL` is a printed string; 022 builds no nullspace and does not back-solve  $\epsilon_{\text{eff}}/Z$  — the `FAIL_TAUTOLOGICAL` firewall, 023’s, is not reintroduced). The fingerprints are dimensionless  $z$ -space rationals; the  $z = a\omega/c_s$  units realization and the residual dimensional gate are 023’s. The  $Z_0^{\text{ret}}/Z_1^{\text{ret}}$  selector (the first concrete Gate-6 input) is the deferred nonlinear solve — 023’s characterized departure.

**Consumed closure and register.** The one *checkable* consumption is Stage-018’s fingerprint  $\{u_2, u_4, v_5\} = \{1/9, 4/81, 1/27\}$ , which 022’s re-derived  $\ell = 2$  fingerprint reproduces (a per-slot / 3e-tooth-backed derive-vs-typed non-regression; the second site is Stage-018’s independently-earned literal,

not a same-machinery reconstruction). *Provenance / context (cited, not re-derived)*: Stage 019/020/021 + the completed pathA\_33 QUAD\_CALIBRATED joint (the prefactor  $P(\omega) = D_0 N / D_{\text{cons}}^2$ , the 54/5 magnitude, the  $\hat{\mu}_0$ -free dimensional closure); Stage 008's raw  $\ell = 0, 1, 2$  outgoing amplitudes; Stage 009/010's bulk Helmholtz mode; and  $c_s$  (the density sound speed, Stage 005 R1  $c_s^2 = 5K\rho^4/m$ ) as the units symbol in  $z = a\omega/c_s$  (distinct from the frozen-wall Helmholtz speed  $c_S$ , 011–017). Register: **zero** new counted knobs; structural edge **R41** (the cross- $\ell$  outgoing-DtN  $-(\ell + 1)/\Lambda_\ell$  fingerprint provenance — the  $\ell = 0, 1, 2$  radiative signature  $\{1, 1/2, 1/27\}$  at  $\{\omega^1, \omega^3, \omega^5\}$  with  $\Lambda_{\text{static}} = -(\ell + 1)$  derived + the scanned order, plus the Gate-4 quadrupole non-regression reproducing the Stage-018 fingerprint component of the now-completed pathA\_33 fold; discharges nothing — earned exterior-wave structure + a consistency non-regression, not a reduction). Part-II counted CALIB set unchanged at  $\{\mu_\eta, T_w, \beta\}$ (013) +  $\{V_{p0}/\ell_c\}$ (015) +  $\{T_\Omega, \beta_2\}$ (017) = 6.

**Verification.** Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed. The .w1 is a genuinely independent route, *re-authored* (not a kept transliteration): it uses Wolfram's built-in `SphericalHankelH1/SphericalHankelH2 + SeriesCoefficient` — a materially different construction from the .py's hand-built `sin/cos jℓ/yℓ + Series/Coefficient` (the source .w1's `branchData` was a near-transliteration of the .py `dtN_branch`, so it was discarded — the mirror-policy transliteration screen). Agreement is at the transcript level (the derived  $\{1, 1/2, 1/27\}$  at  $\{\omega^1, \omega^3, \omega^5\}$ ,  $\Lambda_{\text{static}} = \{-1, -2, -3\}$  with  $h_{\text{mut}} = zh$  firing each and  $H1 \leftrightarrow H2$  shown inert, the scanned order, the  $\ell = 2$  non-regression, the per-slot mutants, the incoming sign-flip, and **3e→FAIL\_QUAD\_REGRESSION**); SymPy 55 PASS / Mathematica 64 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend: Codex folded five blocking findings (the .w1 must be re-authored independent, not the transliterated `branchData`; the normalization and raw-order teeth must be genuinely derived — a  $\Lambda_{\text{static}}$  from the log-derivative + a coefficient scan replacing the source's  $X \equiv X$  set-then-compare and preselected-order check; the  $\ell = 2$  tuple was double-counted and  $\chi_Q$  was subsumed, so the duplicate was removed,  $\chi_Q$  de-counted, and per-slot  $u_2/u_4$  mutants added; the earned/deferred partition is a subset relation with an explicit local/joint output; the checkable consumption is Stage-018's fingerprint), then a Codex confirm-pass caught a further directive-level reasoning error (the proposed  $\Lambda_{\text{static}}$  ablation “outgoing→incoming” is *inert* — both branches give  $-(\ell + 1)$  — so the mutant must be a pole-order corruption  $h_{\text{mut}} = zh$ ; and  $\Lambda_{\text{static}}$ 's subsumed  $\hat{Y}_\ell(0) = 1$  is de-counted). A Grok-4.5 compute-verify of the folded directive returned **DIRECTIVE\_CLEAN**, independently confirming the cross- $\ell$  series, the pole-order-vs-inert mutant, the scan-vs-preselect counterexample, the  $\chi_Q$  subsumption identity, and the  $u_2/u_4$  isolation; a final Codex confirm closed the bookend. Full tri-review on fresh agents (arbiter re-run reproduced the build SymPy 56 / Mathematica 65, exit 0, CWD-independent): **FIDELITY\_CLEAN** (an independent read hand-re-derived the cross- $\ell$  fingerprints, the pole-order  $\Lambda_{\text{static}}$  mutant vs the inert branch swap, the scan-vs-preselect order, the  $\chi_Q$  subsumption, and the per-slot isolation — all computed not stamped, no 019/023 leakage, a genuine built-in `SphericalHankelH1 .w1`) and **ADVERSARIAL\_ISSUES** (both confirm-pass reasoning catches held live-proven, all core physics teeth genuine; two low-severity redundancy teeth — a vacuous **3e rerun\_gate\_logic** constant-length check and a subsumed read-set-exclusion tooth — were remediated, the first make-genuine [compare the two traced verdicts] and the second an honest de-count [the adjacent exact-equality tooth retains the guard], plus two .w1 nits [a symbol typo and a mutation-aware parity fix]), re-verified **REVERIFY\_CLEAN** by the coupling meta-test (each remediated construct fires under a mutation of its object and goes vacuous when neutered; no regression, both engines). Tallies 56/65 → 55/64 (net −1 per engine from the honest de-count). Symbolic per-tooth ablation, mutations on copies.

## Appendix C

# Part III Stage Cards

### C.1 Stage 003: Transverse Photon Pair and the Stray Longitudinal Mode

**Part / anchor.** Part III – Light

**Claim status.** OPEN

**Purpose.** Test whether the shear-surface brane carries light of the Maxwell kind. The transverse question earns a clean result – two massless photons at  $c_\gamma^2 = \mu_R/\rho_{br}$ ; the longitudinal question earns a characterized departure – one stray second-class mode where Maxwell has a gauge-removed zero, with the tuned Maxwell locus reachable only by tuning.

**Status.** OPEN. FAIL-headline stage: the top-line verdict is a **FAIL\_\***, but the earned content is the headline and the FAIL is the first-class characterized landing (no softening). *Earned headline:* the brane carries *light* – two massless transverse photons, with (for the first time in the program)  $c_\gamma^2 = \mu_R/\rho_{br}$ ; verdict **PASS\_TRANSVERSE\_UNDISTURBED**. *Top-line departure:* **FAIL\_CAUCHY\_STRAY\_LONGITUDINAL** – on the provenance-fixed single-medium branch the  $(\nabla \cdot u) + \theta$  sector carries one stray propagating longitudinal DOF, a Dirac–Bergmann *second-class* pair rather than a Maxwell first-class Gauss gauge-removal. *Reachability:* the tuned Maxwell locus ( $K_\theta = C_J^2/\rho_{br}$ ,  $B_{\text{eff}} = 0$ ,  $m_\theta^2 = 0$ ) is reachable and emits **C5\_RESOLVED\_MAXWELL\_BY\_TUNING** – so the FAIL is not hardcoded – but **BY\_TUNING**, not **WITH\_PROVENANCE**.

**Derivation.** The primitive brane Lagrangian (no pre-completed square used as input) is

$$L = \frac{1}{2}\rho_{br}(\partial_t u)^2 - \frac{1}{2}\mu_R(\nabla \times u)^2 - \frac{1}{2}B(\nabla \cdot u)^2 + J(\partial_t \theta) \delta \rho_B + \frac{1}{2}K_\theta(\nabla \theta)^2.$$

**Transverse (earned).** For a polarization  $\perp \mathbf{k}$ ,  $\nabla \cdot u = 0$  and the  $\theta$  couplings vanish, so  $L_T = \frac{1}{2}\rho_{br}(\partial_t u_T)^2 - \frac{1}{2}\mu_R k^2 u_T^2$ , giving

$$\omega^2 = \frac{\mu_R}{\rho_{br}} k^2, \quad c_\gamma^2 = \frac{\mu_R}{\rho_{br}},$$

two polarizations  $\Rightarrow$  `physical_dof` = 2, massless. The speed is able-to-fail:  $\epsilon = 2\rho_{br}$  shifts it to  $\mu_R/(2\rho_{br})$  (**FAIL\_TRANSVERSE\_DISTURBED**). **Longitudinal (departure).** Number conservation slaves  $\delta \rho_B = -\rho_{B0} \nabla \cdot u$ ; integration by parts gives the Maxwell-form cross-term with the *derived* sign  $C_J = -J\rho_{B0}$ , and the finite conjugate-density stiffness the Cauchy modulus  $B_{\text{eff}} = \rho_{B0}^2/\chi_c$ . Since  $\partial_t \theta$  enters

only through the first-order cross-term, the momenta are  $p_u = \rho_{br} \partial_t u_L$ ,  $\pi_\theta = Jk\rho_{B0}u_L$ , forcing the primary constraint  $\Phi_1 = \pi_\theta - Jk\rho_{B0}u_L$ ; preservation gives  $\Phi_2 = -k(Jp_u\rho_{B0} + k\kappa_{\text{phase}}\rho_{br}\theta)/\rho_{br}$ , and

$$\{\Phi_1, \Phi_2\} = \frac{k^2(J^2\rho_{B0}^2 + \kappa_{\text{phase}}\rho_{br})}{\rho_{br}} \neq 0$$

$\Rightarrow$  second-class pair (0 first-class, 2 second-class),  $N_{\text{phys}} = (4 - 2)/2 = 1$ . The finite-stiffness pole  $\omega^2 = k^2\kappa_{\text{phase}}\rho_{B0}^2/[\chi_c(J^2\rho_{B0}^2 + \kappa_{\text{phase}}\rho_{br})]$  has positive residue and a bounded reduced Hamiltonian: a real stray longitudinal mode, not a ghost and not a gauge-removal.

**Physical use, fidelity, and the departure.** The brane carries light: two transverse photons at  $c_\gamma^2 = \mu_R/\rho_{br}$  from the medium's own moduli (light = in-plane brane shear). Honest scope: the order-parameter phase  $\theta$  does *not*, on its frozen definitions, supply a gauge-removed Maxwell scalar – it leaves one stray second-class longitudinal mode; the obstruction is pinned to the Cauchy modulus  $B_{\text{eff}} = \rho_{B0}^2/\chi_c$  plus the conventional phase-gradient sign  $K_\theta = -\kappa_{\text{phase}} < 0$  (vs the electric sign  $K_\theta = C_J^2/\rho_{br} > 0$ ). Fidelity: the decisive sign  $C_J = -J\rho_{B0}$  is *derived* here by an in-script integration by parts (the earned source had asserted it), closing that stage's known **FIDELITY\_ISSUES**; the conventional  $K_\theta$  sign is a labeled postulate whose verdict-dependence is exercised by the reachable Maxwell-locus control. Resolution of the stray mode is calibrated / sim-deferred;  $\mu_R, \rho_{br}$  are calibrated inputs.



## Appendix D

# Reproducibility Map

### D.1 Purpose of This Appendix

Placeholder for the rebuilt ledger's reproducibility rules.

### D.2 Reproducibility Layers

| Layer                                    | What it controls | Minimum evidence required |
|------------------------------------------|------------------|---------------------------|
| No reproducibility layers populated yet. |                  |                           |

### D.3 Canonical Source Registry

Placeholder for source artifacts.

### D.4 Primary Symbolic-Audit Registry

Placeholder for symbolic audit artifacts.

### D.5 Repository Freeze Protocol

Placeholder for future release-freeze requirements.

## Appendix E

# Source File Index

### E.1 Purpose of This Appendix

Placeholder for the rebuilt ledger’s source-file index.

### E.2 Source Classes

Placeholder for source classes.

### E.3 Generated Ledger Files

| File family                                   | Role                              |
|-----------------------------------------------|-----------------------------------|
| <code>paper/frontmatter/*.tex</code>          | Placeholder governance chapters.  |
| <code>paper/parts/*.tex</code>                | Future theorem-block chapters.    |
| <code>paper/stages/stage_NNN.tex</code>       | Future per-stage cards.           |
| <code>scripts/ledger_stageNNN_*.py</code>     | Future SymPy audit scripts.       |
| <code>mathematica/ledger_stageNNN_*.wl</code> | Future Mathematica audit scripts. |

### E.4 Precedence and Conflict-Resolution Rules

Placeholder for source precedence and conflict-resolution rules.

## Appendix F

# Layer-2 Fit-Insertion Index

### F.1 Generated Coverage Summary

| Item               | Count |
|--------------------|-------|
| Phase-A candidates | 0     |
| Audited candidates | 0     |

### F.2 Phase-A Candidate Rows

No candidate rows exist in the empty ledger skeleton.

### F.3 Visible Grouping and Exclusion Accounting

No grouping or exclusion accounting exists yet.

## Appendix G

# Layer-2 Parameter-Provenance Audit

### G.1 Generated Coverage Summary

| Item                                   | Count |
|----------------------------------------|-------|
| Top-level provenance parameter records | 0     |
| Canonical candidates                   | 0     |

### G.2 Published-Target Benchmark Sources

No benchmark rows exist in the empty ledger skeleton.

### G.3 Parameter Genealogy Rows

No parameter genealogy rows exist yet.

### G.4 Visible Alias Accounting

No alias accounting exists yet.

## Appendix H

# Fill Workflow and Editorial Rules

### H.1 Purpose of This Appendix

Placeholder for the rebuilt ledger fill workflow.

### H.2 What Filled Means

Placeholder for completion criteria.

### H.3 Governing Maps Used During Filling

Placeholder for status, notation, anchor, dependency, source, and reproducibility controls.

### H.4 Stage Appendix Fill Protocol

Placeholder for future stage-card authoring rules.

### H.5 Theorem-Block Fill Protocol

Placeholder for future Part authoring rules.

### H.6 Minimal Build and Static-Check Workflow

1. Confirm every retained `\input` target exists.
2. Run the SymPy and Mathematica audit runners.
3. Build `paper/pde_ledger.tex`.